
TECHNICAL SUPPORTING AND REQUIREMENTS DOCUMENT

Non-Confidential Technical Supporting and Requirements Document: Localised Safety-Oriented Battery Management Platform for Commercial EV Integration

Government Support and Stage-Gated Validation Proposal

Applicant identity: **Barak Microelectronics**

Submission route: Tamil Nadu Technology Transfer and Facilitation Centre (TNTTFC) / iTNT Hub
technology-transfer portal

Prepared for Indian government and quasi-government technical reviewers assessing technical support, shared test infrastructure, expert mentorship, stage-gated research and development funding, pre-compliance guidance, manufacturing readiness and pilot facilitation for an MSME startup undertaking full-stack battery-management-system design and development.

Version 1.0 | 11 August 2026

Legal-entity particulars, promoter details, partner arrangements and letters of intent are deliberately not asserted in this document and must be supported by dated documentary evidence before filing.

Contents

1. Document Control, Classification and Reader Guidance

- 1.1 Document control
- 1.2 Classification and permitted use
- 1.3 Intended audience
- 1.4 Status of this document
- 1.5 Claim-status legend
- 1.6 How to read this document

2. Executive Summary

- 2.1 Evidence-status box
- 2.2 What is being asked for
- 2.3 What will be delivered
- 2.4 What is expressly not claimed

3. Product Definition, Public-Interest Need and Government Additionality

- 3.1 Product definition
- 3.2 Problem statement
- 3.3 Beneficiaries
- 3.4 Government additionality

4. Current Development Status and Evidence Baseline

- 4.1 Maturity statement
- 4.2 Evidence inventory
- 4.3 Evidence-boundary statements the applicant makes affirmatively

5. Intended Use, Reference Configuration, System Boundary and Scope

- 5.1 Intended use
- 5.2 Initial commercial-EV reference configuration
- 5.3 System boundary
- 5.4 In scope and out of scope for the funded programme

6. Black-Box Capability Overview

7. Requirements Register

- 7.1 Requirement record format

8. Sanitized Architecture Narrative and External Interfaces

- 8.1 Public architecture description
- 8.2 Rationale for the reference component category
- 8.3 External interfaces

9. Safety and Regulatory Pathway

- 9.1 Precise terminology
- 9.2 Applicable regulatory route
- 9.3 Staged compliance plan
- 9.4 EMC, cybersecurity and quality roadmap

10. Verification and Validation Strategy

- 10.1 Evidence ladder
- 10.2 Evidence gates G0–G5
- 10.3 Test-result graduation rules
- 10.4 Minimum validation matrix
- 10.5 Validation of predictive and diagnostic outputs
- 10.6 Traceability matrix structure
- 10.7 Key acceptance principles

11. Work-Package Plan (12–18 Months)**12. Expected Project Results and Long-Term Product Outcomes**

- 12.1 Programme deliverables (what public support buys)
- 12.2 Long-term product outcomes (beyond the programme)
- 12.3 Key performance indicators

13. Commercial Viability — Conservative Framing

- 13.1 Beachhead rationale
- 13.2 Customer qualification stages
- 13.3 Pricing
- 13.4 Market and revenue
- 13.5 Fleet interface and data

14. Government Support Requested

- 14.1 Portal submission fit
- 14.2 Support package requested
- 14.3 Eligible cost categories
- 14.4 Multiple funding asks are different capital envelopes
- 14.5 Milestone-linked disbursement principles

15. Localisation, DVA and Scheme-Linkage Evidence Plan

- 15.1 Wording control
- 15.2 DVA evidence plan
- 15.3 Wider policy context

16. Risk Register**17. Confidentiality, Disclosure Model and Rights**

- 17.1 Three-tier disclosure model
- 17.2 Protected material excluded from all public disclosure
- 17.3 Rights statement
- 17.4 What government reporting will and will not include
- 17.5 Document handling guidance

18. Claims Register: High-Risk Source Wording and Approved Replacements**19. Submission-Readiness Checklist and Conclusion**

- 19.1 Accompanying items for the portal application
- 19.2 Pre-submission checks
- 19.3 Conclusion

Appendix A. Source Register**Appendix B. Abbreviations**

1. Document Control, Classification and Reader Guidance

1.1 Document control

Field	Entry
Document title	Non-Confidential Technical Supporting and Requirements Document: Localised Safety-Oriented Battery Management Platform for Commercial EV Integration
Subtitle	Government Support and Stage-Gated Validation Proposal
Applicant / product identity	Barak Microelectronics (trading identity used consistently across the applicant's own technical corpus). Legal-entity particulars, incorporation status, promoter details, registered address and authorised signatory are deliberately not asserted in this document and must be evidenced before filing.
Intended submission route	Tamil Nadu Technology Transfer and Facilitation Centre (TNTTFC) / iTNT Hub technology-transfer portal, under its stated "Prototype to Product" startup pathway. ¹⁴
Applicant category asserted	Micro / Small / Medium Enterprise startup. The portal's Innovation Voucher application accepts Micro, Small and Medium Enterprises and DPIIT-certified startups and requires a UAM / MSME or Startup India registration certificate. ¹⁶ The applicant must attach the certificate that actually exists; no registration number is asserted here.
Version / date	Version 1.0 — 11 August 2026
Document owner	Technical lead, Barak Microelectronics
Classification	Public / Non-Confidential — Proprietary algorithms and methodology intentionally excluded
Validity	This issue is valid for six months from the date of issue or until superseded by a controlled revision, whichever is earlier.
Revision history	1.0 — first controlled issue prepared for technology-transfer and deep-tech support evaluation.

1.2 Classification and permitted use

This is a public, non-confidential dossier. It contains no source code, algorithms, equations, model structures, feature definitions, calibration data, thresholds, decision rules, excitation or sensing parameters, fault-signature libraries, security keys, schematics, layout data or complete bill of materials. Every technical statement in this document is written at black-box level: observable inputs, observable outputs, external interfaces, target requirements, verification methods and evidence artifacts.

The recipient may use this document solely to evaluate the requested technical and financial support. No licence, assignment or other right in any intellectual property is granted by its disclosure. A formal three-tier disclosure model and rights statement are set out in Section 17.

1.3 Intended audience

- Technology-transfer and facilitation officers assessing a startup's prototype-to-product pathway and knowledge-partner needs.¹⁴
- Deep-tech programme officers assessing technology risk, public value and stage-gated funding fit.⁸
- Independent technical reviewers and mentors in functional safety, embedded electronics, battery systems, cybersecurity and validation.
- Test-infrastructure and laboratory coordinators assessing shared-facility utilisation.
- Compliance and standards advisors assessing the regulatory pathway for a traction-battery electronic subsystem.

14. TNTTFC / iTNT Hub — Technology Transfer and Facilitatio... — <https://www.techtransfer.itnthub.tn.gov.in/>

16. TNTTFC — Innovation Voucher Programme application (Vouc... — <https://www.techtransfer.itnthub.tn.gov.in/innovation-voucherform/>

8. iTNT Hub — Tamil Nadu Deep Tech Startup Policy (nodal a... — <https://itnthub.tn.gov.in/deeptechpolicy.html>

1.4 Status of this document

This document is a support-seeking technical dossier for an early-stage research and development programme. It is not a product datasheet, a certification claim, a type-approval document, a safety case, or an investment prospectus. It does not represent any figure in it as a demonstrated product result unless that figure is explicitly labelled Demonstrated and traced to a dated test artifact — and, at the date of issue, no such artifact exists.

1.5 Claim–status legend

Every substantive assertion in this document carries one of four labels. The labels are used consistently in requirement tables, outcome tables and the claims register.

Label	Meaning	Permitted wording	Present coverage in this dossier
Demonstrated	A reproducible result supported by a dated test artifact identifying configuration, protocol, instrumentation, data and acceptance result.	"Demonstrated in test report ."	None. No physical test artifact exists at the date of issue.
Planned	An approved intention, requirement or design decision that has not yet been proved.	"The programme will...", "the design is intended to..."	Most architecture, process and governance statements.
Target	A quantified pass criterion for a future test, stated with its configuration and reference method.	"Target: ..., subject to validation under protocol ."	All performance figures.
Projection	A commercial, socio-economic or market proposition requiring further evidence.	"Projected", "scenario", "to be evaluated"	All commercial and economic figures.

1.6 How to read this document

Sections 2 to 5 establish need, status and scope. Section 6 is the technical core and contains the requirements register. Sections 7 to 10 describe architecture, regulatory pathway, verification strategy and the work-package plan. Sections 11 to 16 describe expected results, outcomes, commercial framing, the support request and the localisation evidence plan. Sections 17 to 20 cover confidentiality, claim control, submission readiness and conclusion. Appendix A is the full source register.

2. Executive Summary

Barak Microelectronics proposes a stage-gated research and development programme to convert a documented, locally engineered battery management system (BMS) concept into a controlled, independently evidenced A-sample research demonstrator for commercial electric-vehicle battery-pack integration. Support is sought through the Tamil Nadu Technology Transfer and Facilitation Centre, whose startup pathway is explicitly described as "Prototype to Product" and which offers Masterclass, IP Clinic, Innovation Spotlight, Research2Revenue, Technology Transfers, IP Management, Spinoff Support, Seed Grant, TRL Validation and Technology Portfolio mechanisms.¹⁴¹⁵

The applicant is an MSME startup undertaking **full-stack BMS design and development**. In this dossier "full-stack" denotes the breadth of the product-development lifecycle and of the externally observable system layers the applicant intends to own — system requirements, electronics platform integration, embedded software, safety and protection functions, diagnostics, external interfaces, cybersecurity, verification and validation, manufacturing readiness, pack and vehicle integration support, an optional governed fleet interface, and commercialisation readiness. It does not denote, and this document does not disclose, any proprietary method, model, calibration or decision rule inside those layers.

2.1 Evidence–status box

Question a reviewer will ask	Honest answer at date of issue
Is there a physical prototype?	No. No physical prototype is evidenced in the applicant's document set.
Is there a formal product certification?	No. No certification, type approval or conformity assessment has been obtained.
Is there a field pilot or OEM qualification?	No. No pilot, purchase order, letter of intent or OEM qualification is represented.

14. TNTTFC / iTNT Hub — Technology Transfer and Facilitatio... — <https://www.techtransfer.itnthub.tn.gov.in/>

15. TNTTFC — Startup programmes: Masterclass, IP Clinic, In... — <https://www.techtransfer.itnthub.tn.gov.in/startup/>

Question a reviewer will ask	Honest answer at date of issue
Is there a confirmed domestic value addition (DVA) certificate?	No. No costed bill of materials has been audited and no DVA certificate exists.
Is there an independently demonstrated predictive-performance result?	No. Predictive diagnostic performance is a validation objective, not a result.
What does exist?	A documented problem analysis, a preliminary system and architecture concept, a candidate commercially available component category selection, a defined initial use case, and a structured validation and work-package plan.
What is the maturity claim?	Applicant-assessed early research stage, approximately TRL 2–3, subject to formal assessment. Formal TRL will be confirmed through an inception evidence review (Gate G0).

2.2 What is being asked for

The applicant requests a package of technical and financial support rather than an undifferentiated grant: milestone-linked early R&D; funding; time-bounded access to shared test infrastructure; independent expert mentorship in functional safety, cybersecurity and validation; pre-compliance and standards-scoping guidance; IP protection guidance; manufacturing-readiness support; and facilitated, safety-gated introductions to pack integrators for later controlled evaluation. These map directly onto published portal mechanisms, including the Innovation Voucher Programme's eligible cost categories and its option to request an iTNT knowledge partner.¹⁶

2.3 What will be delivered

At programme close the applicant will deliver an A-sample research demonstrator; a frozen external interface package; an approved requirements and traceability baseline; bench and fault-injection evidence; an independent technical review record; a pre-compliance gap roadmap; a localisation and DVA evidence plan; a manufacturing-readiness plan; and a documented pilot-readiness go/no-go decision. The programme will not deliver, and does not promise, vehicle homologation, product certification, production readiness or a demonstrated safety guarantee.

2.4 What is expressly not claimed

Current limitations and non-claims. The project is at an early design stage. No claim is made that the proposed product is presently certified, type-approved, production-ready, proven to prevent battery fires, validated to replace physical safety sensors, or demonstrated to achieve any stated performance target in a vehicle. Use of automotive safety-capable components does not by itself establish product-level ISO 26262 compliance. Projected local value addition, scheme relevance, price, cost savings, revenue and socio-economic effects are planning assumptions subject to independent verification. Any advanced diagnostic output will supplement, and will not bypass, conventional protection until supported by an approved safety case.

3. Product Definition, Public-Interest Need and Government Additionality

3.1 Product definition

The product is an embedded battery-pack supervisory and protection subsystem for commercial electric vehicles. It measures pack and cell conditions through approved sensing hardware, maintains independent protection functions, produces state and health outputs with validity information, exchanges defined information with the vehicle and charger through a controlled network interface, records diagnostic evidence, and optionally exports non-safety health summaries to a governed fleet interface. All estimation and diagnostic processing is intended to run locally on the vehicle so that essential monitoring and protection do not depend on network connectivity.

16. TNTTFC — Innovation Voucher Programme application (Vouc... — <https://www.techtransfer.itnthumb.tn.gov.in/innovation-voucherform/>)

3.2 Problem statement

- **Dependence on imported traction-battery electronics.** A substantial share of BMS content in Indian commercial EV packs is imported. Import dependence concentrates design authority, calibration knowledge, field-data interpretation and spare-part continuity outside India. The applicant treats the precise import share as an estimate requiring a cited source before publication.
- **Operating-condition mismatch.** Electronics and calibration developed for temperate duty cycles are not necessarily representative of Indian commercial fleet operation — high ambient temperature, high daily energy throughput, opportunity charging and mixed maintenance practice. This is a hypothesis the programme is designed to test with instrumented data, not an established quantified result.
- **Safety evidence, not raw incident counts.** The Ministry of Road Transport and Highways has stated in Parliament that centralised data tracking battery-fire incidents nationally is not maintained.¹³ The public-interest argument is therefore made on severity and evidence grounds rather than incident frequency: a single passenger-carrying commercial-vehicle battery event has extreme consequence, and the regulatory response — the AIS-156 and AIS-038 (Rev. 2) amendments — already defines the compliance floor.¹
- **Localisation direction of travel.** National schemes increasingly reward or require domestic manufacture of power-electronics assemblies. PM E-DRIVE's phased manufacturing programme for e-buses and e-trucks has been amended to tighten localisation of power-electronics assemblies,¹¹¹² and PLI-Auto applies a 50% domestic value-addition condition to eligible approved products.⁴⁷ A locally engineered BMS is aligned with that direction; it does not by itself confer eligibility under either scheme.
- **Evidence-generation gap for small firms.** The binding constraint on a small Indian electronics team is not idea generation but access to calibrated high-voltage benches, cell simulation, environmental and EMC pre-compliance capability, fault-injection capability and independent functional-safety review.

3.3 Beneficiaries

Beneficiary	Benefit sought (all Planned / Target)
Commercial EV pack integrators and assemblers	A locally supported, interface-documented BMS option with accessible engineering support and a defined evidence pack for their own pack-level approval work.
Commercial fleet operators	Better-evidenced condition information for maintenance planning; local service and diagnostic support; no dependence on external connectivity for protection.
Vehicle OEMs	An additional domestic supply option for a safety-relevant subsystem, with configuration control and traceability suited to their qualification processes.
Test agencies and laboratories	Additional, well-prepared domestic demand for validation and pre-compliance services, with reusable, documented test protocols.
MSME electronics suppliers in Tamil Nadu	Qualification opportunities in PCB assembly, harnesses, enclosures, fixtures and calibration services under an automotive-quality discipline.
Engineering workforce	Skilled roles and structured training in battery safety, embedded systems, verification, cybersecurity and quality.
Public interest	Retention of design authority, calibration knowledge and field data for Indian commercial-vehicle battery systems within India.

3.4 Government additionality

Additionality is the test of whether public support changes the outcome. The applicant states plainly what will not happen, or will happen materially more slowly and with lower evidence quality, in the absence of support:

- Access to calibrated high-voltage benches, cell simulation, thermal, vibration and EMC pre-compliance facilities would not be obtainable at the required standard; scarce capital would be diverted from evidence generation into equipment procurement.
- Independent functional-safety, cybersecurity and validation review would be deferred, which is precisely the review that prevents an early-stage safety-relevant product from accumulating undetected design debt.

13. PIB — MoRTH eDAR reply on EV accident and fire incident... — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2225866>

1. PIB / MoRTH — Amendments to EV battery safety standards... — <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1862596®=3&lang=2>

11. PIB — PM E-DRIVE scheme: outlay, e-bus support and phas... — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2290696®=3&lang=1>

12. <https://auto.economictimes.indiatimes.com/news/commercial-vehicle/government-amends-localisation-norms-for-e-buses-e-trucks-under-pm-e-drive-scheme/124268731>

4. PIB — PLI Scheme for Automobile and Auto Components: ou... — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2287124®=3&lang=1>

7. PIB — PLI Scheme for Automobile sector: 50% domestic va... — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2290691®=3&lang=1>

- Standards-scoping with a competent test agency would occur too late, creating a material risk of spending on the wrong test scope.
- Structured IP protection guidance would be deferred past the point at which public capability descriptions have already been circulated.
- Controlled, insured, safety-gated engagement with a pack integrator would not be achievable by an unsupported small firm.

Support is requested as catalytic and co-funded. The applicant does not represent that private capital is unavailable or unnecessary; it represents that public support materially accelerates the generation of independent evidence that private capital and customers both require before they will engage.

4. Current Development Status and Evidence Baseline

4.1 Maturity statement

The applicant assesses the project as being at an early research stage — approximately TRL 2–3, subject to formal assessment. This is an applicant self-assessment, not a validated determination. A formal TRL determination will be produced at the inception evidence review (Gate G0) and, if the programme is supported, may be confirmed through the portal's TRL Validation mechanism.¹⁴

The evidence boundary is explicit. A TRL 3 claim ordinarily requires analytical or experimental proof-of-concept output that can be inspected. The applicant's document set contains a documented concept, a preliminary architecture and a component-category rationale, but does not contain an executed simulation study report, a dataset, a bench log or any other inspectable experimental artifact. Accordingly the applicant claims no more than “approximately TRL 2–3, subject to formal assessment”, and treats the production of a reviewable model-level study as the first funded evidence deliverable rather than as an accomplished fact.

The applicant does not claim TRL 3 or TRL 4 as a fact, and does not claim TRL 6 or above. The portal states that Research2Revenue Seed Fund eligibility applies to startups at TRL 6+.¹⁵ The applicant is not presently eligible on that basis and does not apply for it in this dossier. R2R is presented as a later gate, to be revisited only after the evidence described in Sections 9 to 11 exists and has been independently reviewed.

4.2 Evidence inventory

Artifact class	Status at date of issue	Consequence for this dossier
Problem analysis and market context	Documented; secondary sources	Usable as context; each external statistic must carry a cited primary source.
System and architecture concept	Documented at concept level	Usable as Planned architecture description; not a validated design.
Component-category selection	Preliminary; commercially available automotive-oriented parts identified	Usable at public component-category level; specific part revisions remain configuration-controlled and are not published here.
Model-level (MIL) study report	Not available	First funded evidence deliverable (WP1).
Bench or HIL test logs	Not available	WP2 deliverable.
A-sample hardware	Not available	WP3 deliverable.
Requirements traceability matrix	Structure defined in this document; not yet populated with results	WP0/WP1 deliverable.
Functional-safety work products (HARA, safety concept)	Not available	WP4 deliverable.
Cybersecurity risk assessment	Not available	WP4 deliverable.
Independent review record	Not available	Requested as in-kind support; WP2/WP4 deliverable.

14. TNTTFC / ITNT Hub — Technology Transfer and Facilitatio... — <https://www.techtransfer.itnthub.tn.gov.in/>

15. TNTTFC — Startup programmes: Masterclass, IP Clinic, In... — <https://www.techtransfer.itnthub.tn.gov.in/startup/>

Artifact class	Status at date of issue	Consequence for this dossier
Pre-compliance or accredited test reports	Not available	WP5 deliverable.
Costed BOM and DVA calculation	Not available	WP6 deliverable; see Section 15.
Customer LOI, MoU, pilot agreement or purchase order	Not available	No customer or partner commitment is represented anywhere in this document.
Legal-entity, incorporation and signatory particulars	Not asserted in this document	Must be evidenced from registry documents before filing.

4.3 Evidence–boundary statements the applicant makes affirmatively

- No physical prototype exists.
- No formal product certification, conformity assessment or type approval exists.
- No field pilot has been conducted and no OEM qualification has been achieved.
- No DVA certificate exists and no audited domestic value-addition percentage is claimed.
- No independently demonstrated predictive-performance result exists.
- No partner, investor, laboratory or customer is represented as committed. Where a relationship is under discussion, it is described as targeted or under discussion, and any letter of intent, memorandum of understanding or facility-access letter must be supported by a dated executed document before it is cited in a filing.

5. Intended Use, Reference Configuration, System Boundary and Scope

5.1 Intended use

The intended use for the funded programme is a single, defined commercial-EV battery-pack integration case: a lithium iron phosphate (LFP) traction pack in an M-category bus or N-category light commercial vehicle application, operating in Indian ambient conditions under commercial fleet duty cycles, with the BMS integrated by a pack assembler or vehicle OEM. Selecting one case is deliberate: the applicant's earlier material spans two-wheeler to high-voltage bus applications, and that breadth is not defensible as a current scope.

5.2 Initial commercial–EV reference configuration

The reference configuration below is a **proposed configuration baseline for the A-sample research programme**. It will be frozen at Gate G0 and thereafter changed only under formal change control. It is stated at the level a reviewer needs to judge scope and test cost, and no further.

Parameter	A-sample reference baseline (Planned)
Vehicle category	M or N category commercial vehicle (bus / light commercial vehicle) application context.
Cell chemistry family	LFP prismatic cells, single defined supplier and model for the A-sample test matrix.
Pack architecture	Modular series-connected string with distributed cell-monitoring front ends and a central supervisory controller.
Voltage class	High-voltage traction class as defined in the frozen configuration record; a single class only. Multi-class scalability is a roadmap statement, not a Phase 1 scope.
Balancing strategy	Passive balancing is proposed as the conservative A-sample architecture decision. This is an architecture decision proposed for approval at G0, not a demonstrated capability. Active balancing is explicitly future scope subject to a separate trade study.

Parameter	A-sample reference baseline (Planned)
Controller platform	An automotive-oriented, commercially available multi-core microcontroller family with lockstep-capable processing, hardware watchdog and memory-protection features. The S32K3 family is named only at public component-category level as the reference architecture for the A-sample. Other processor options remain roadmap or trade-study items and are not baselined.
Cell-monitoring front end	Commercially available automotive cell-monitoring devices with an isolated daisy-chain communication layer.
Vehicle interface	CAN / CAN-FD with a released interface control document and DBC; diagnostics over a standard automotive diagnostic service protocol.
Connectivity	Optional, non-safety fleet gateway. Disabled by default in the bench and any shadow-mode phase.

5.3 System boundary

Boundary element	Definition
Inputs	Cell and pack voltage measurements from approved sensing hardware; pack current measurement; physical temperature measurements; authorised vehicle and charger commands; configurable pack parameters; authenticated diagnostic service requests; optional fleet-gateway inputs (non-safety).
Outputs	Pack status; permitted charge and discharge state and limits; derating requests; warning and fault status with validity information; state and health estimates with validity and uncertainty indication; diagnostic and event records; authorised vehicle-network messages; optional non-safety fleet health summaries.
Safe-state objective	On a detected or diagnosed condition requiring protection, the system requests transition to the defined restricted or isolated state as allocated by the integrated pack and vehicle safety concept.
External dependencies (not the product)	Cell selection and qualification; pack mechanical and thermal design; contactors, fuses and pre-charge hardware; charger and vehicle control unit behaviour; enclosure and harness; telematics hardware; cloud application; and all pack-level and vehicle-level verification.

5.4 In scope and out of scope for the funded programme

In scope (funded programme)	Out of scope (explicitly excluded)
Requirements baseline, configuration control and traceability	Vehicle homologation and type approval
Supervisory controller and monitoring-interface integration at A-sample level	Formal ISO 26262 assessment or certification
Embedded software for monitoring, protection, state output and diagnostics	Battery cell development or cell qualification
Bench and fault-injection verification, including connectivity-loss behaviour	Pack mechanical, thermal-barrier and enclosure design
Functional-safety and cybersecurity planning work products	Charger, DC-DC converter and traction-inverter design
Pre-compliance screening and standards applicability scoping	Series production, tooling and volume manufacturing
Manufacturing-readiness and supplier-qualification planning	Commercial cloud platform development or data monetisation
Localisation and DVA evidence method definition	Any field deployment with control authority over a vehicle
Pilot-readiness decision preparation	Marketing, sales inventory and general corporate overhead

6. Black-Box Capability Overview

The capabilities below describe what the system is intended to do and what a reviewer can observe and test from outside the product. They deliberately omit how each capability is realised. Every entry is Planned or Target; none is Demonstrated.

#	Capability (observable)	Observable output / interface	Label	How it will be judged
C1	Synchronous monitoring of the defined pack variables across the configured string	Periodic status frames with per-channel validity	Planned	Bench measurement against calibrated reference instrumentation
C2	Independent protection response to defined abnormal conditions	Protection state and isolation request on the vehicle interface	Planned	Fault-injection tests with timestamped traces
C3	State-of-charge output with declared uncertainty and validity	Numeric output plus validity and uncertainty fields	Target	Comparison against a traceable reference procedure over a defined test matrix
C4	State-of-health / degradation indicator output	Numeric indicator plus validity	Target	Aged-cell and reference-method comparison
C5	Supplementary internal thermal-state indication	Supplementary diagnostic output, clearly flagged as an estimate	Target	Cross-validation against instrumented reference cells; never a replacement for physical over-temperature protection in the A-sample safety concept
C6	Early-warning indication for defined abnormal-condition classes	Warning score, warning state, affected channel, confidence and timestamp	Target	Per-fault-class validation with sensitivity, false-alarm rate and lead-time distribution reported separately
C7	Adaptive charge and discharge limit management	Limit and derating messages to charger and vehicle controller	Planned	Limit-response and interface tests
C8	Cell balancing within the frozen A-sample strategy	Balancing status and termination reporting	Planned	Bench balancing, thermal and efficiency measurement
C9	Connectivity-independent operation of essential protection	Continued local protection with the external network removed	Planned	Connectivity-loss test trace
C10	Forensic diagnostic and event recording	Readable event records with configuration and version identity	Planned	Readout inspection and retention-policy review
C11	Authenticated service, configuration and update control	Access-controlled diagnostic and update procedures	Planned	Security and update-control test evidence
C12	Optional governed fleet health interface	Non-safety summary export, disabled by default	Planned	Data-flow review and separation demonstration

Boundary of the proprietary element. The system contains a proprietary on-device estimation and diagnostic module. This dossier describes only its inputs, its outputs, its validity and confidence reporting, its degraded-mode behaviour, and the way its outputs will be verified. Its internal method, model class, parameters, calibration and decision rules are outside the scope of any public disclosure and are not reconstructed anywhere in this document.

7. Requirements Register

7.1 Requirement record format

Each requirement carries a stable identifier, a type, a single “shall” statement, a rationale, a priority, a verification method, an acceptance criterion with the evidence artifact that will demonstrate it, a status label and a confidentiality classification. Requirements are written as observable behaviour and interface contracts. No requirement embeds a proprietary method. Numeric acceptance limits that would reveal protection thresholds or tuning are held in the controlled design record and are referenced here by protocol identity rather than by value.

Priority values are M (must, in scope for the A-sample), S (should, in scope if the gate schedule permits) and R (roadmap, out of scope for the funded programme). Confidentiality values are P (public), N (available under NDA to a named evaluator) and X (restricted; not disclosed).

7.2 Stakeholder and business requirements

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
STK-001 Stakeholder	The programme shall deliver an A-sample research demonstrator with a released external interface package suitable for evaluation by a pack integrator. <i>Rationale:</i> integrators cannot evaluate a subsystem without a frozen interface contract.	M Planned P	Design review and inspection. <i>Evidence:</i> A-sample dossier and released interface control document.
STK-002 Stakeholder	The programme shall produce an evidence pack that allows an independent reviewer to reproduce the reported results. <i>Rationale:</i> public support purchases evidence, not narrative.	M Planned P	Independent review. <i>Evidence:</i> protocol register, raw-data retention record, reviewer memorandum.
STK-003 Business	The programme shall establish a costed bill of materials and localisation model sufficient to support a later domestic value-addition assessment under the applicable methodology. <i>Rationale:</i> localisation claims must be auditable before they are made.	M Planned N	Analysis and financial review. <i>Evidence:</i> costed BOM, supplier-origin records, calculation method note.
STK-004 Business	The applicant shall maintain a claims register in which every published quantitative claim is traceable to a labelled evidence status. <i>Rationale:</i> prevents recurrence of the inconsistencies present in earlier internal drafts.	M Planned P	Document audit. <i>Evidence:</i> claims register, version-controlled.
STK-005 Business	The applicant shall retain design authority, engineering records and field-data interpretation capability within India for the supported programme. <i>Rationale:</i> retention of capability is a principal public-interest justification.	M Planned P	Inspection of records and IP register. <i>Evidence:</i> IP register and engineering-record index.

7.3 System, functional and measurement requirements

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
SYS-001 System	The system shall support the single initial commercial-EV pack configuration recorded in the frozen configuration baseline. <i>Rationale:</i> scope control; prevents untestable multi-segment claims.	M Planned P	Inspection and bench test. <i>Evidence:</i> approved configuration record.
SYS-002 System	The system shall retain independent protection functions when higher-level estimation or diagnostic functions are unavailable, invalid or degraded. <i>Rationale:</i> defence in depth; advanced functions must never be the sole protection.	M Planned P	Fault injection and design review. <i>Evidence:</i> degraded-mode test trace and safety analysis extract.

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
SYS-003 System	The system shall identify all external dependencies and non-product items in the released documentation. <i>Rationale:</i> prevents attribution of pack-level or vehicle-level responsibility to the subsystem.	M Planned P	Inspection. <i>Evidence:</i> boundary and assumptions-of-use document.
MEA-001 Measurement	Cell voltage, pack voltage, pack current and physical temperature inputs shall each meet a defined accuracy, resolution, update rate, synchronisation and calibration requirement recorded in the requirements baseline. <i>Rationale:</i> every downstream output inherits measurement quality.	M Target P	Bench test against calibrated reference instruments. <i>Evidence:</i> measurement verification report with instrument calibration certificates.
MEA-002 Measurement	Measurement validity and stale-data status shall be visible to every downstream function and on the external interface. <i>Rationale:</i> silent stale data is a classic latent failure in monitoring systems.	M Planned P	Fault injection. <i>Evidence:</i> validity-propagation test trace.
MEA-003 Measurement	Any value derived by estimation rather than direct measurement shall be explicitly identified as an estimate and shall report validity and confidence. <i>Rationale:</i> integrators must not mistake an estimate for a measurement.	M Planned P	Interface inspection and bench test. <i>Evidence:</i> interface control document and output trace.
MEA-004 Measurement	Measurement channel diagnostics shall detect open-circuit, short-circuit, out-of-range and implausible-value conditions on safety-relevant channels. <i>Rationale:</i> sensing faults must be distinguishable from real pack conditions.	M Planned P	Fault injection. <i>Evidence:</i> channel-diagnostic test matrix and results.

7.4 Estimation and health-output requirements

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
EST-001 Performance	The system shall output pack state of charge with a validity flag and an uncertainty measure at the defined update rate throughout the approved operating envelope. <i>Rationale:</i> an estimate without declared uncertainty cannot be used safely by an integrator.	M Target P	Test against a traceable reference procedure over the defined matrix. <i>Acceptance target:</i> state-of-charge error not exceeding 2% as an A-sample target under the defined test matrix (chemistry, temperature band, duty cycle, ageing state and reference method specified in the protocol). Tighter performance is a stretch objective only. <i>Evidence:</i> SOC verification report with maximum error and distribution, not averages alone.
EST-002 Performance	The system shall output a state-of-health / degradation indicator with defined units, bounds, update rate, initialisation behaviour, recovery after reset and degraded mode. <i>Rationale:</i> health outputs drive maintenance decisions and must be defined before they are trusted.	M Target P	Bench test with new and aged cells. <i>Evidence:</i> SOH characterisation report including out-of-distribution behaviour.
EST-003 Performance	The system shall provide a supplementary internal thermal-state estimate that shall be cross-validated against instrumented reference cells and shall not replace independent physical over-temperature protection unless and until supported by an approved safety case. <i>Rationale:</i> removal of physical sensing must be earned by evidence.	M Target P	Instrumented-cell comparison. <i>Evidence:</i> thermal-estimate validation report and validity envelope.

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
EST-004 Performance	For each specified abnormal-condition class the system shall report a warning score, warning state, affected channel, confidence status and timestamp; sensitivity, false-alarm rate and lead-time distribution shall be reported per class and shall not be pooled. <i>Rationale:</i> pooled early-warning claims are not falsifiable.	M Target P	Controlled fault-scenario validation. <i>Acceptance target:</i> an early-warning objective of at least 48 hours applies only to the defined scenario classes named in the protocol and is pending validation; it is not a general product claim. <i>Evidence:</i> per-class validation report with confusion matrix and lead-time distribution.
EST-005 Performance	Vehicle range prediction, if ever offered, shall be treated as a separate vehicle-level function with separate data, separate acceptance criteria and separate verification. <i>Rationale:</i> range accuracy and SOC accuracy are different metrics and were conflated in earlier drafts.	R Planned P	Requirements review. <i>Evidence:</i> scope statement in the requirements baseline.
EST-006 Performance	Loss or invalidity of any advanced estimation or diagnostic function shall cause a defined degraded mode and shall not disable hard protection. <i>Rationale:</i> the single most important safety property of an analytics-bearing controller.	M Planned P	Fault injection. <i>Evidence:</i> degraded-mode trace and hazard linkage record.

7.5 Safety, protection and timing requirements

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
SAF-001 Safety	The system shall monitor the required pack and cell conditions and shall communicate a defined protection state to the pack and vehicle integration layer when a configured protection condition is reached. <i>Rationale:</i> the primary protection contract with the integrator.	M Planned P	HIL and bench fault injection. <i>Evidence:</i> timestamped test log with configuration identity and response trace.
SAF-002 Safety	The system shall detect loss of a safety-relevant measurement or communication path and shall enter the allocated safe or degraded state within the approved timing budget. <i>Rationale:</i> loss of observability is itself a hazard.	M Planned P	Fault injection. <i>Evidence:</i> trace, pass/fail record and hazard linkage.
SAF-003 Safety	Independent protection coverage shall be specified for over-voltage, under-voltage, over-temperature, under-temperature as applicable, over-current and short-circuit interfaces, isolation loss, communication loss, processor fault, supply fault and implausible sensor data. <i>Rationale:</i> complete protection coverage must be enumerated, not assumed.	M Planned P	Design review and fault injection. <i>Evidence:</i> protection coverage matrix with test references.
SAF-004 Safety	The allocation of safety responsibility among the BMS, the contactor driver, the charger, the vehicle control unit and the pack shall be documented and agreed with the integrator. <i>Rationale:</i> unallocated responsibility is the most common integration failure.	M Planned P	Joint review with integrator. <i>Evidence:</i> allocation table and assumptions-of-use document.
SAF-005 Timing	Detection latency, protection command-issuance latency and physical isolation completion shall be specified, budgeted and verified as three separate quantities. <i>Rationale:</i> electronic command issuance and electromechanical contactor opening operate on different time scales; conflating them produces a physically challengeable claim.	M Target P	Instrumented timing measurement. <i>Acceptance:</i> an electronic detection-and-command budget in the microsecond class and a separate mechanical isolation budget in the millisecond class, both per the frozen contactor specification; values recorded in the controlled timing budget. <i>Evidence:</i> timing verification report with oscilloscope traces.

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
SAF-006 Safety	A predictive or advisory warning shall never suppress, delay or relax a hard protection threshold. <i>Rationale:</i> analytics must be additive to safety, never subtractive.	M Planned P	Design review and fault injection. <i>Evidence:</i> architecture review record and negative-test results.
SAF-007 Safety	The final pack integration shall maintain an independent protection path such that a single identified controller or software fault does not defeat the allocated safety goal. <i>Rationale:</i> single-point-failure avoidance at pack level.	M Planned P	Design review, dependent-failure analysis and fault injection. <i>Evidence:</i> safety analysis and independent review record.

7.6 Balancing, power-limit, communications and diagnostics requirements

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
BAL-001 Functional	The A-sample shall implement the frozen balancing strategy — proposed as passive balancing — with defined activation constraints, maximum balancing current, thermal limits, termination criteria and fault handling. <i>Rationale:</i> the conservative baseline matches the A-sample scope and budget; active balancing is future scope.	M Planned P	Bench test with thermal measurement. <i>Evidence:</i> balancing performance and thermal report.
PWR-001 Interface	The system shall publish charge and discharge power-limit information with a defined update rate, timeout behaviour and plausibility checking, and the behaviour of the vehicle controller and charger on message loss shall be specified. <i>Rationale:</i> limit messages are safety-relevant control inputs to other systems.	M Planned P	Interface and timeout tests. <i>Evidence:</i> interface conformance report.
COM-001 Interface	The system shall exchange defined status, diagnostic and control information through the agreed vehicle-network interface in accordance with a released interface control document. <i>Rationale:</i> integration contract.	M Planned P	Network conformance test. <i>Evidence:</i> DBC / ICD release and interoperability log.
COM-002 Interface	Message timing, end-to-end protection, timeout handling and ownership of the interface definition shall be specified and placed under configuration control. <i>Rationale:</i> uncontrolled interface drift breaks integrator qualification.	M Planned P	Inspection and bus-error injection. <i>Evidence:</i> ICD baseline and error-injection results.
DIA-001 Diagnostics	The system shall record selected fault events, time markers and configuration and version identity for authorised post-event analysis. <i>Rationale:</i> forensic evidence supports both field safety and the integrator's approval dossier.	M Planned P	Bench test and inspection. <i>Evidence:</i> readout sample and retention policy.
DIA-002 Diagnostics	Diagnostic sessions shall define access levels, audit logging and calibration permissions. <i>Rationale:</i> uncontrolled calibration access is a safety and liability exposure.	M Planned P	Security review and test. <i>Evidence:</i> access-control test record.
OPS-001 Functional	Essential local monitoring, protection, fault logging and safe-state request shall remain available without any external network connection. <i>Rationale:</i> offline autonomy is a core design property and a public-interest resilience argument.	M Planned P	Connectivity-loss test. <i>Evidence:</i> test trace demonstrating continued local protection.

7.7 Cybersecurity, update and change-control requirements

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
SEC-001 Cyber-security	Only authenticated and authorised service procedures shall be able to modify safety-relevant configuration or deploy firmware. <i>Rationale:</i> unauthorised configuration change is a direct safety hazard.	M Planned P	Security test and review. <i>Evidence:</i> access-control and update-control evidence.

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
SEC-002 Cyber-security	The product shall implement secure boot, signed firmware, controlled debug-port access, least-privilege service access and secure logging. <i>Rationale:</i> baseline automotive security hygiene for a network-connected controller.	M Planned P	Security test. <i>Evidence:</i> security verification report (summary public; detail restricted).
SEC-003 Cyber-security	A threat analysis and risk assessment appropriate to the vehicle interface shall be completed before any external connectivity is enabled. <i>Rationale:</i> connectivity must follow analysis, not precede it.	M Planned N	Independent review. <i>Evidence:</i> TARA report and review minutes.
SEC-004 Cyber-security	Key provisioning, rotation and revocation, credential handling and a documented vulnerability-handling process shall be defined; a software bill of materials shall be maintained. <i>Rationale:</i> lifecycle security, supply-chain transparency and incident readiness.	M Planned N	Process audit. <i>Evidence:</i> key-management procedure, SBOM, vulnerability process record.
OTA-001 Change control	Any field-update capability shall verify update integrity and shall preserve an approved recovery and rollback path; over-the-air update shall remain disabled for safety-relevant functions until independently validated. <i>Rationale:</i> an update mechanism is itself a hazard until proven.	M Planned P	Test and design review. <i>Evidence:</i> update test report and recovery demonstration.
OTA-002 Change control	No in-field learning or unreviewed model or calibration update shall occur for any safety-relevant function. <i>Rationale:</i> uncontrolled adaptation invalidates the verification evidence.	M Planned P	Design review. <i>Evidence:</i> architecture statement and configuration-management plan.
CFG-001 Change control	Every firmware, model and calibration release shall be versioned, reproducible, signed, traceable to its test evidence and rollback-controlled. <i>Rationale:</i> without release traceability no test result can be attributed to a product state.	M Planned P	Configuration audit. <i>Evidence:</i> release register and build reproducibility record.

7.8 Environmental, reliability, manufacturing, serviceability, data and integration requirements

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
ENV-001 Environmental	Operating and storage temperature, humidity, vibration, shock, ingress protection at enclosure level, electromagnetic compatibility, power transients, electrostatic discharge, design life, duty cycle and dormant current shall each be specified in the requirements baseline. <i>Rationale:</i> environmental scope must precede test budgeting.	M Target P	Pre-compliance screening and, later, accredited testing. <i>Evidence:</i> environmental and EMC pre-compliance reports.
ENV-002 Reliability	Component datasheet ratings shall be recorded separately from validated assembly performance, and no component rating shall be published as a product-level result. <i>Rationale:</i> component capability is not system evidence.	M Planned P	Document audit. <i>Evidence:</i> ratings-versus-results register.
MFG-001 Manufacturing	Each production-intent unit shall carry a unique traceability identifier linked to its approved hardware revision, software release, end-of-line test result and critical component lot data. <i>Rationale:</i> traceability is a precondition for any field-safety action.	M Planned P	Process audit. <i>Evidence:</i> traceability record and end-of-line test template.
MFG-002 Manufacturing	An approved-vendor list, incoming-inspection method, counterfeit-avoidance control, programming and key-injection procedure, calibration records, serialisation and nonconformance control shall be defined before any pilot build. <i>Rationale:</i> manufacturing quality cannot be retrofitted after a pilot.	S Planned N	Quality-plan review. <i>Evidence:</i> supplier-quality plan and DFM/DFT review record.

ID / Type	Shall statement and rationale	Priority / Status / Class	Verification and evidence artifact
SRV-001 Serviceability	Field service, repair policy, return analysis, spare-part and obsolescence policy shall be defined for the pilot configuration. <i>Rationale:</i> commercial fleets buy serviceability, not only hardware.	S Planned P	Review. <i>Evidence:</i> service and obsolescence plan.
DAT-001 Data governance	Fleet data collection, if enabled, shall be opt-in and contractual, purpose-limited, access-controlled, retention-bounded and functionally separated from safety-critical operation. <i>Rationale:</i> data governance must precede any data feature.	M Planned P	Review and demonstration. <i>Evidence:</i> data-flow map, contractual clause, access logs.
DAT-002 Data governance	No telemetry shall be used for model improvement, transfer, sale or disclosure beyond the contracted purpose without documented authorisation and applicable legal review. <i>Rationale:</i> secondary use without authority is both a legal and a trust failure.	M Planned P	Process audit. <i>Evidence:</i> data-management plan and authorisation records.
DAT-003 Data governance	Safety functions shall not depend on the availability of any dashboard, gateway or cloud service. <i>Rationale:</i> restates OPS-001 as a data-governance constraint.	M Planned P	Connectivity-loss test. <i>Evidence:</i> test trace and architecture statement.
INT-001 Integration	The applicant shall provide the integrator with an assumptions-of-use document listing every condition the pack or vehicle must satisfy for the subsystem's stated behaviour to hold. <i>Rationale:</i> the honest counterpart to a subsystem safety claim.	M Planned P	Joint review. <i>Evidence:</i> assumptions-of-use document countersigned by the integrator.
INT-002 Integration	The applicant shall supply structured evidence to support the integrator's pack-level or vehicle-level approval work and shall not represent that evidence as an approval of the subsystem. <i>Rationale:</i> a BMS contributes to, and cannot substitute for, integrated approval.	M Planned P	Documentation review with integrator and test agency. <i>Evidence:</i> integration evidence pack index.

8. Sanitized Architecture Narrative and External Interfaces

8.1 Public architecture description

At public disclosure level the architecture is a conventional, well-understood automotive pattern. Cell and pack measurements are acquired by distributed automotive cell-monitoring front ends across a galvanically isolated communication layer and are delivered to a central supervisory controller built on an automotive-oriented multi-core microcontroller with lockstep-capable processing, memory protection and hardware watchdog features. The controller hosts three logically separated concerns: a measurement and validation layer; a proprietary estimation and diagnostic module, shown publicly only as a labelled black box with declared inputs and outputs; and an independent safety manager that evaluates raw and validated measurements against protection criteria and issues protection and isolation requests without depending on the proprietary module. External interfaces expose status, limits, warnings, diagnostics and optional non-safety health summaries.

The public data-flow statement is therefore: **cell and pack inputs → measurement and validation → proprietary estimation and diagnostics (black box) → independent safety manager → contactor, vehicle-controller and charger interfaces → optional non-safety fleet gateway**. No public diagram or narrative in this or any related public document will show excitation or sensing parameters, waveform or frequency characteristics, model families or topology, feature or parameter flows, thresholds, execution partitioning, task scheduling, timing internals or circuit paths.

8.2 Rationale for the reference component category

The A-sample is built on commercially available automotive components whose safety-relevant features are documented by their suppliers in publicly available safety documentation. This choice is disclosed because a reviewer legitimately needs to know that the safety argument rests on qualified automotive silicon rather than on general-purpose parts. It is disclosed only at component-category level. The applicant states expressly that supplier component qualification is not system certification, and that supplier feature claims will be verified against the exact part number, revision, safety manual and integration assumptions actually used.

8.3 External interfaces

Interface	Direction	Public description	Control status
Cell-monitoring link	Input	Isolated daisy-chain link to distributed cell-monitoring front ends	Configuration-controlled; electrical detail restricted
Pack current measurement	Input	Pack-level current sensing input with diagnostics	Configuration-controlled
Physical temperature inputs	Input	Discrete physical temperature sensing channels retained in the A-sample safety concept	Baselined
Vehicle network (CAN / CAN-FD)	Bidirectional	Status, limits, warnings, protection state and control information per released ICD and DBC	ICD under configuration control; released to integrator
Diagnostic service interface	Bidirectional	Authenticated diagnostic sessions, event readout and authorised calibration	Access levels defined; seed/key and credential logic restricted
Contactor / isolation interface	Output	Protection and isolation request to the pack's contactor control path	Allocation agreed with integrator
Charger interface path	Bidirectional	Charge permission, limits and derating information	Per integration agreement
Service and programming interface	Bidirectional	Controlled programming and provisioning access, disabled in field configuration	Restricted
Optional fleet gateway	Output	Non-safety, opt-in, purpose-limited health summary export; disabled by default	Governed under DAT-001 to DAT-003

9. Safety and Regulatory Pathway

9.1 Precise terminology

The dossier distinguishes six statements that are frequently and incorrectly treated as interchangeable. The applicant can presently make only the first.

#	Statement	Applicant status
1	Selected components are designed for use in safety-related applications and are supported by supplier safety documentation.	Applicable, as a statement about suppliers' products, not about the applicant's product.
2	The product architecture is intended to support an ISO 26262-aligned safety lifecycle using safety-capable components.	This is the applicant's design intent. Formal assessment is future work.
3	The development process follows selected ISO 26262 work products.	Planned from WP0; not yet evidenced.
4	An independent gap assessment has been completed.	Not achieved. Requested as supported activity.
5	The product or system has been assessed or certified to a named scope.	Not achieved and not claimed.
6	The battery pack / REESS or the vehicle holds type approval under the applicable AIS scope.	Not achieved, not claimed, and not the applicant's to hold.

The applicant does not use the phrase “ASIL-D certified”, or any equivalent, in relation to this project. The correct formulation, used throughout this dossier, is: **architecture intended to support an ISO 26262-aligned safety lifecycle using safety-capable components; formal assessment is future work.**

9.2 Applicable regulatory route

Applicability depends on vehicle category. Government clarification identifies AIS-156 as the standard for L-category electric power-train vehicles, and AIS-038 (Rev. 2) as the corresponding traction-battery standard for M and N category vehicles.² Both standards were brought into force through phased amendments with effect from December 2022 and March 2023.¹ Because the initial use case is a bus or light commercial vehicle, **AIS-038 (Rev. 2) is the likely applicable traction-battery route, not AIS-156.** AIS-156 applies to L-category vehicles and is referenced here only for completeness.

Two further points are stated to prevent a common category error. First, the applicable standard describes approval of the vehicle and/or the rechargeable electrical energy storage system, supported by submitted technical specifications, with relevant changes treated as controlled modifications.³ There is no standalone BMS certificate in this framework. The correct description of the applicant's contribution is **planned contribution to the integrated pack and vehicle approval evidence.** Second, final applicability — including vehicle category, pack boundary, test scope, EMC and environmental requirements — must be confirmed in writing with the pack integrator and a notified test agency before any formal test expenditure is committed.

The BMS supports integrated pack and vehicle approval evidence. It cannot by itself guarantee prevention of thermal runaway, and it cannot by itself confer type approval. Thermal-runaway outcomes depend on cell selection, pack thermal and mechanical design, containment, charging control, vehicle design and operating procedure, none of which are within the product boundary defined in Section 5.

9.3 Staged compliance plan

Stage	Claimable statement at that stage	Evidence artifact	Responsible body
Now (concept)	Architecture intended to support an ISO 26262-aligned lifecycle using safety-capable components; no assessment performed.	Safety plan outline; component safety documentation review	Applicant
Gate 0 — applicability	Vehicle category, pack boundary and applicable standard scope confirmed in writing.	Written scoping note or quotation from a competent agency; standards applicability matrix	Applicant with notified test agency and integrator
Gate 1 — safety planning	Hazard analysis and risk assessment, safety goals and functional safety concept established.	HARA, safety goals, functional safety concept, preliminary FMEA/FTA, assumptions of use, competence plan	Applicant with independent mentor
Gate 2 — design evidence	Technical safety concept implemented with traceability and tested safety mechanisms.	Technical safety concept, traceability matrix, fault-injection report, hardware-software interface specification, cybersecurity TARA	Applicant with independent reviewer
Gate 3 — pre-compliance	Independent architecture gap analysis completed; EMC and environmental pre-checks performed; corrective actions closed.	Gap-analysis report, pre-compliance reports, corrective-action closure record	Qualified assessor and laboratory
Gate 4 — integrated approval support	Evidence submitted in support of the integrator's pack or vehicle approval, for the agreed configuration and scope only.	Integration evidence pack; agency-agreed scope statement	Pack integrator / vehicle OEM with notified agency
Later (post-programme)	Formal functional-safety assessment for a defined item and configuration, if and when funded and justified.	Assessment report from a competent independent assessor	Independent assessor

2. PIB / MoRTH — Clarification on battery safety standards... — <https://www.pib.gov.in/PressReleaseDetail.aspx?PRID=2202970®=1&lang=1>

1. PIB / MoRTH — Amendments to EV battery safety standards... — <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1862596®=3&lang=2>

3. MoRTH — AIS-156 official standard text (approval of veh... — <https://morth.nic.in/sites/default/files/ASI/AIS-156.pdf>

9.4 EMC, cybersecurity and quality roadmap

- **EMC and environmental.** AIS-004 and the associated EMC and environmental requirements form part of the integrated approval context. The programme funds pre-compliance screening only; formal testing follows the agency-agreed scope at a later gate.
- **Cybersecurity.** A threat analysis and risk assessment, secure boot and signed update controls, key management, SBOM maintenance and a vulnerability-handling process are planned work products (SEC-001 to SEC-004). No cybersecurity certification is claimed.
- **Quality.** A staged quality roadmap is planned: configuration management and change control from WP0; design-review and traceability discipline from WP1; supplier-quality, incoming-inspection and end-of-line test method from WP6. Automotive quality-management-system certification is a post-programme objective, not a programme deliverable.
- **Public procurement.** No public-procurement eligibility or preference is claimed. Procurement readiness will be assessed only after product, quality, approval and seller-registration prerequisites are met.

10. Verification and Validation Strategy

10.1 Evidence ladder

Public support should purchase evidence rather than components. The programme therefore sequences work so that each stage produces an inspectable artifact before the next stage is funded.

1. Requirements and architecture baseline, under configuration control.
2. Feasibility study of the candidate measurement and processing path, with resource and power budgets.
3. Model-level (MIL) study with versioned datasets and a documented study design.
4. Software-level (SIL) testing with static analysis and coverage reporting.
5. Hardware-in-the-loop testing with systematic fault injection.
6. Instrumented cell and module bench testing against calibrated references.
7. A-sample verification against the requirements baseline.
8. Independent review and laboratory pre-compliance screening.
9. Shadow-mode vehicle observation with no control authority — only if a gate authorises it.
10. Controlled authority-enabled pilot — only after a safety gate, integrator and insurer consent and independent review.
11. Industrialisation and the applicable approval programme, outside this programme's scope.

10.2 Evidence gates G0–G5

Gate	Purpose	Minimum entry evidence	Exit deliverable	Decision
G0 — Baseline	Establish a controlled project and an honest maturity baseline.	Product boundary, frozen use case and configuration, owner, risk register, initial requirements, TRL evidence review.	Requirements baseline, configuration record, confidentiality and IP plan, TRL assessment note.	Authorise bench setup.
G1 — Bench bring-up	Show the instrumented demonstrator functions safely under controlled conditions.	Asset list, laboratory safety plan, test-jig qualification, calibration status.	Dated bring-up report, measurement traceability record, basic interface test.	Authorise the test campaign.
G2 — Fault-injection evidence	Prove allocated protective behaviour against selected injected faults.	Approved protocols, pass criteria, frozen configuration baseline.	HIL and bench results, anomaly log, independent reviewer note.	Decide A-sample maturity.

Gate	Purpose	Minimum entry evidence	Exit deliverable	Decision
G3 — A-sample readiness	Confirm the physical A-sample and interface package are ready for an integrator.	Revised design, DFM review, pre-check plan for EMC and environmental screening.	A-sample dossier, released ICD, safety-plan update, manufacturing traceability plan.	Authorise controlled pack integration work.
G4 — Pack-level pre-compliance	Evaluate the integrated representative pack against the relevant plan.	Integrator agreement, pack configuration, notified-agency engagement.	Pre-compliance report and corrective-action plan.	Decide pilot suitability.
G5 — Pilot readiness	Authorise limited supervised activity only if risks are closed to the agreed level.	Training, field monitoring plan, incident plan, cybersecurity and data controls, insurance and contract terms.	Pilot protocol and documented go/no-go record.	Pilot or return to development.

10.3 Test-result graduation rules

A claim may graduate from Target or Projection to Demonstrated only when all six of the following exist. Absent any one of them, the claim remains a Target.

1. The exact product configuration and firmware or build identifier.
2. An approved test protocol with the environmental and operating range stated, pre-registered before execution.
3. Calibrated instrumentation and a traceable reference method.
4. Retained raw or source data with an auditable analysis record.
5. A defined acceptance criterion and the actual measured result, including maximum error and distribution rather than averages alone.
6. Reviewer sign-off, independent where the claim is safety-relevant or commercially material.

10.4 Minimum validation matrix

Every performance test is stratified by: cell supplier and model; chemistry and format; new and aged state; state-of-charge band; ambient and cell temperature; charge and discharge rate; rest and dynamic duty cycles; sensor tolerance and communication-fault conditions; pack or module topology; and firmware and model version. Every report states sample size and exclusion rules, reference instruments and calibration status, protocol identity and pre-registration date, maximum error and distribution, confidence intervals, false positives and false negatives with missed-event severity, repeatability and reproducibility, limitations and out-of-distribution behaviour, and raw-data retention and independent-access arrangements.

10.5 Validation of predictive and diagnostic outputs

This is the highest-risk claim family in the entire dossier and is therefore governed by its own protocol. Each abnormal-condition class is defined and validated independently; the event constituting a critical failure is defined per class; warning-threshold governance is documented in the restricted design record without publishing threshold values; sensitivity, specificity, false alarms per operating hour and per vehicle-month, and both median and lower-bound lead time are reported per class; the consequence of a missed event is stated; the operator action associated with each warning is defined; fail-safe behaviour when the diagnostic is unavailable or uncertain is specified; and each test states whether the fault was simulated, induced or naturally occurring. Results from different fault classes are never pooled into a single lead-time claim.

10.6 Traceability matrix structure

Column	Content
Requirement ID	Stable identifier from Section 7.
Source	Stakeholder need, standard clause, test-agency requirement or internal objective.
Hazard / risk link	Reference to the hazard analysis entry, where safety-relevant.

Column	Content
Design element	Architectural element or work product implementing the requirement (restricted detail held in the design record).
Verification method	Inspection, analysis, test, demonstration or independent assessment.
Protocol ID	Identifier of the approved test protocol.
Evidence artifact	Report, raw-data reference, trace, review minutes or certificate.
Result and date	Pass, fail, partial or not-yet-executed, with date and build identity.
Status label	Demonstrated, Planned, Target or Projection.
Confidentiality	Public, NDA or restricted.

10.7 Key acceptance principles

- Advanced analytics are defence in depth, never the sole protection.
- Physical sensing and independent hardware protection remain in the baseline until their removal is justified by a complete, approved safety case.
- Every analytic output carries plausibility checks, validity and confidence flags, watchdog supervision, stale-data handling and a defined safe fallback.
- No performance figure is published without a test-report identifier.
- Safety limits, thresholds and decision rules remain in the restricted design record and never appear in a public document.
- A failed gate is a legitimate programme outcome that triggers re-scoping, not a result to be re-narrated.

11. Work–Package Plan (12–18 Months)

The plan below is structured as work packages with objective exit gates and named evidence deliverables. Durations are indicative and will be rebaselined at G0 against confirmed facility availability and component lead times.

WP	Months	Objective and principal activities	Exit gate	Evidence deliverables
WP0 Governance and baseline	0–1	Freeze the use case, reference configuration and balancing decision; establish requirements baseline, configuration management, risk register, confidentiality and IP plan, data-management plan; conduct the inception TRL evidence review; complete standards-applicability scoping enquiry.	G0 approved	Frozen configuration record; requirements baseline v1; TRL assessment note; risk register; IP and confidentiality plan; standards-scoping enquiry record.
WP1 Feasibility and modelling	1–3	Component and measurement-path trade study with supplier engineering support; resource, memory, timing and power budgets; model-level study with versioned datasets; test strategy and protocol register.	Feasibility review passed	Trade-study report; resource-budget note; MIL study report with dataset versions; protocol register v1; candidate BOM structure.
WP2 Bench capability and core functions	3–6	Qualified bench and test jig; instrumented cell string; bring-up of monitoring, protection, limit management, balancing and interface functions; systematic fault injection; connectivity-loss testing.	G1 and G2 passed	Bring-up report; calibration and traceability record; fault-injection report with anomaly log; connectivity-loss trace; interface conformance log; independent reviewer note.
WP3 A-sample and interface package	6–9	A-sample hardware revision under controlled release; manufacturing and calibration records; end-of-line test definition; released interface control document; verification against the requirements baseline.	G3 passed	A-sample dossier; released ICD and DBC; verification report against requirements baseline; manufacturing traceability concept; DFM/DFT review record.

WP	Months	Objective and principal activities	Exit gate	Evidence deliverables
WP4 Safety and cybersecurity work products	4–10 (parallel)	HARA and safety goals; functional and technical safety concepts; FMEA and FTA; dependent-failure considerations; TARA; secure boot, signed update and authenticated diagnostic testing; configuration and incident-response plans.	Independent review completed	HARA; safety concept documents; FMEA/FTA; TARA; security test summary; configuration-management and incident plans; independent review minutes.
WP5 Pre-compliance and approval scoping	9–13	Written scoping with a competent test agency; EMC, environmental and vibration pre-compliance screening; independent architecture gap analysis; corrective-action closure; revised cost and schedule for later formal testing.	G4 readiness decision	Agency scope letter or quotation; pre-compliance reports; gap-analysis report; corrective-action closure record; revised compliance cost estimate.
WP6 Manufacturing readiness and localisation evidence	10–15	Supplier shortlist and qualification criteria; costed BOM and origin records; localisation model and DVA calculation method; quality plan; end-of-line test method; obsolescence and dual-sourcing policy.	Manufacturing-readiness review	Costed BOM (restricted); supplier scorecards; DVA method note and calculation; quality plan; EOL test template.
WP7 Pilot-readiness preparation	13–18	Shadow-mode pilot protocol design; integrator and operator engagement under written terms; monitoring, incident-response, insurance and data-control arrangements; go/no-go criteria definition and decision.	G5 go/no-go recorded	Pilot protocol; contract and insurance framework; monitoring and incident plan; documented pilot-readiness decision record.
WP8 Programme management and dissemination	0–18 (continuous)	Gate reviews, reporting, evidence-repository maintenance, claims-register maintenance, mentor engagement, IP clinic participation, publication of the non-confidential requirements template.	All gate reports accepted	Gate review minutes; evidence repository index; claims register; published non-confidential requirements and responsible-validation template.

12. Expected Project Results and Long-Term Product Outcomes

12.1 Programme deliverables (what public support buys)

These are the results the programme itself will produce and against which it should be judged. Each is an artifact that can be inspected by a reviewer.

#	Programme deliverable	Acceptance basis
1	A-sample research demonstrator for the frozen reference configuration	Physical unit, controlled hardware revision, verification report against the requirements baseline
2	Frozen external interfaces	Released interface control document and DBC under configuration control, accepted in a joint review
3	Requirements and traceability baseline	Populated traceability matrix with verification method, protocol identity and status per requirement
4	Bench and fault-injection evidence	Dated test reports with configuration identity, calibration records, anomaly log and retained raw data
5	Independent technical review	Signed review minutes from a reviewer independent of the design team, covering safety, cybersecurity and claim integrity
6	Pre-compliance gap roadmap	Gap-analysis report against the agency-confirmed applicable scope, with closure actions and revised cost estimate
7	DVA and localisation evidence plan	Documented calculation method, costed BOM structure, supplier-origin evidence procedure
8	Manufacturing-readiness plan	Supplier qualification criteria, quality plan, end-of-line test method, traceability concept, DFM/DFT review
9	Pilot-readiness go/no-go decision	Documented decision record against pre-defined criteria, including a legitimate no-go outcome
10	Public, non-confidential requirements and validation template	Published artifact reusable by other Indian teams working on safety-relevant battery electronics

12.2 Long-term product outcomes (beyond the programme)

These are outcomes the applicant intends to pursue after, and conditional upon, the programme's evidence. They are stated as Projections and are explicitly not programme deliverables.

- Safer diagnostics: earlier, better-characterised indication of defined abnormal conditions in commercial packs, supporting maintenance intervention before escalation.
- Uptime and maintenance support: condition information that supports planned maintenance rather than reactive replacement in commercial fleets.
- Offline autonomy: protection behaviour that does not degrade when connectivity is unavailable, which matters in Indian intercity and depot operations.
- Local engineering and IP retention: design authority, calibration knowledge and field-data interpretation retained within Tamil Nadu and India.
- Test-ecosystem utilisation: sustained, well-prepared demand for domestic validation and pre-compliance services, with reusable protocols.
- Skilled employment: durable roles in functional safety, embedded systems, validation, cybersecurity and quality.
- MSME supplier readiness: local suppliers qualified to automotive quality discipline for assembly, harnesses, enclosures and fixtures.
- Import-risk reduction: reduced exposure to single-source foreign supply for a safety-relevant subsystem.
- Data sovereignty: Indian commercial-fleet battery telemetry governed under Indian contractual and legal control rather than embedded in foreign-controlled closed systems.

12.3 Key performance indicators

Baselines below are the honest position at date of issue. Targets are stated as target types with measurement sources, not as invented numbers. Numeric job, revenue and import-substitution figures are deliberately not asserted.

KPI	Baseline (today)	Project-end target type	Measurement source	Caveat
Approved test protocols executed	Zero	A defined, gate-linked count of executed protocols agreed at G0	Protocol register and dated test reports	Count depends on facility availability
Independent reviews completed	Zero	At least one independent safety and cybersecurity review per major gate	Signed review minutes	Depends on mentor availability
Requirements verified against baseline	Zero	A stated percentage of must-priority requirements verified	Traceability matrix	Verification of some requirements is deferred to pack level
Fault classes validated	Zero	A defined set of abnormal-condition classes validated per class	Per-class validation reports	Per-class results are not poolable
Shared test-facility sessions used	Zero	A booked and utilised session count	Laboratory booking and utilisation logs	Subject to facility allocation
Local suppliers qualified	Zero	A shortlist qualified against defined criteria	Supplier scorecards	Qualification is not a purchase commitment
Engineers trained in safety, validation, cyber and quality	Not quantified	A defined number of engineers and interns with recorded training outcomes	Training records and certificates	Training capacity depends on programme funding
IP protection actions	No formal register	IP register established; filing or trade-secret decisions recorded	IP register and decision minutes	Filing decisions require counsel input
Localisation evidence maturity	No costed BOM	Costed BOM and documented DVA calculation method complete	Costed BOM and method note	A DVA percentage is not claimed at programme end
Public knowledge artifact	None	One published non-confidential requirements and validation template	Published document	Excludes all proprietary content

KPI	Baseline (today)	Project-end target type	Measurement source	Caveat
Integrator technical evaluations	Zero	A defined number of structured technical evaluations initiated	Evaluation records	An evaluation is not an order or a qualification

13. Commercial Viability — Conservative Framing

Commercial content is included only to the extent a public reviewer needs it to judge whether the technology has a realistic route out of the laboratory. It is not a fundraising case, and no figure in this section is a commitment.

13.1 Beachhead rationale

The initial focus is commercial-EV pack integration — buses and light commercial vehicles — for four reasons that can be tested rather than asserted: the consequence of a battery event in a passenger-carrying or high-utilisation asset is severe, which raises the value of evidence-grade diagnostics; commercial operators quantify downtime, so maintenance-planning information has measurable value; pack assemblers in this segment are more accessible to a small supplier than passenger-car OEM programmes; and national localisation policy attention is currently concentrated on this segment.¹¹¹² Each of these is a hypothesis to be tested through structured customer discovery during the programme.

13.2 Customer qualification stages

The applicant will report customer engagement only against the following ladder, and will not describe any relationship at a stage higher than the documentary evidence supports.

Stage	Definition	Evidence required
Target account	Identified as a potential integrator or operator	Internal list only; no external representation
Contacted	Initial contact made	Dated correspondence
Discovery completed	Structured technical problem interview completed	Interview record
Technical evaluation agreed	Both parties agree to a defined technical evaluation	Written scope agreement
NDA signed	Mutual confidentiality agreement executed	Executed agreement
LOI / MoU	Non-binding intent recorded	Executed dated document
Paid development contract	Funded engineering work commissioned	Executed contract
Purchase order	Order placed	Executed order

At the date of issue the applicant is at the **target account** stage. No letter of intent, memorandum of understanding, evaluation agreement, contract or purchase order exists, and no named organisation is represented as a partner, customer, investor or endorser.

13.3 Pricing

Unit pricing is treated as a provisional financial-model assumption by configuration and volume, not as a published price. The applicant's earlier internal drafts contain materially different price points for the same nominal product; rather than select one of them for a government document, this dossier states the principle: a price will be derived from the costed BOM produced in WP6, plus verified assembly, test and support costs, and will be presented by configuration and by volume band with the underlying assumptions disclosed. No price is asserted here.

11. PIB — PM E-DRIVE scheme: outlay, e-bus support and phas... — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2290696®=3&lang=1>

12. <https://auto.economictimes.indiatimes.com/news/commercial-vehicle/government-amends-localisation-norms-for-e-buses-e-trucks-under-pm-e-drive-scheme/124268731>

13.4 Market and revenue

Market size and revenue appear only as scenario planning in a separate financial annex maintained outside this technical dossier. The applicant's earlier drafts contain mutually inconsistent market series and revenue paths; none is carried forward. Any future market statement will name one publisher, one geography, one segment definition and one date, and will not be mixed with a differently defined series. Revenue is not a programme deliverable and is not used to justify the support request.

13.5 Fleet interface and data

The optional fleet dashboard is a non-safety, governed, opt-in support function. It is deliberately de-emphasised: it is not a monetisation case, it is not part of the safety argument, and it is disabled by default in the bench and any shadow-mode phase. Data-monetisation intent is excluded from this dossier entirely. Any future data service would require documented customer authority, purpose limitation, retention limits, access control, de-identification review and legal review before it could be offered.

14. Government Support Requested

14.1 Portal submission fit

This dossier is prepared for submission through the TNTTFC / iTNT technology-transfer portal, whose startup pathway is described as “Prototype to Product”.¹⁴ The matrix below maps the applicant's needs onto published portal mechanisms. Where the applicant is not currently eligible, that is stated.

Portal mechanism	Applicant fit	What is requested
Prototype to Product pathway ¹⁴	Direct fit. The programme is precisely a documented concept moving toward an evidenced prototype.	Admission to the pathway with gate-linked oversight.
Innovation Voucher Programme — Voucher A (up to Rs.3 lakh) and Voucher B (up to Rs.7 lakh) ¹⁶	Direct fit for MSME / DPIIT-startup applicants. The applicant will apply under the classification it can actually evidence.	Support against the published eligible cost categories listed in Section 14.3.
Request for an iTNT knowledge partner ¹⁶	Direct fit. Independent functional-safety, validation and cybersecurity expertise is the applicant's largest capability gap.	Assignment of a knowledge partner for safety, validation and compliance review.
IP Clinic ¹⁵	Direct fit. Trade-secret and filing sequencing decisions are live and time-sensitive.	One-to-one IP protection and filing-strategy sessions.
IP Management and Technology Transfer / JIGSAW network ^{15,17}	Partial fit. The applicant seeks access to academic instrumentation, characterisation expertise and researcher collaboration rather than inbound technology licensing.	Facilitated introductions to institutional laboratories and researchers for characterisation and independent review.
TRL Validation ¹⁴	Direct fit and explicitly welcomed. The applicant asks to have its own maturity claim tested rather than accepted.	Independent TRL assessment at G0 and re-assessment at G3.
Masterclass ¹⁵	Partial fit for commercialisation and licensing literacy.	Participation where scheduling permits.
Innovation Spotlight ¹⁵	Later-stage fit only. The applicant does not seek visibility before it has evidence.	Deferred until after G3.
Research2Revenue Seed Fund (TRL 6+) ¹⁵	Not currently eligible. The applicant assesses itself at approximately TRL 2–3 and does not claim TRL 6+.	Presented as a later gate, to be revisited only after independent evidence supports a higher TRL.
Seed Grant / Spinoff Support ¹⁴	Not applied for in this dossier.	No request made.
Deep-tech policy instruments (nodal agency: iTNT Hub) ^{8,9}	Contextual fit for TRL-banded early R&D; support and shared test beds.	Considered as a complementary envelope; see Section 14.4.

14. TNTTFC / iTNT Hub — Technology Transfer and Facilitatio... — <https://www.techtransfer.itnthub.tn.gov.in/>

16. TNTTFC — Innovation Voucher Programme application (Vouc... — <https://www.techtransfer.itnthub.tn.gov.in/innovation-voucherform/>

15. TNTTFC — Startup programmes: Masterclass, IP Clinic, In... — <https://www.techtransfer.itnthub.tn.gov.in/startup/>

17. iTNT Hub — Technology Transfer / JIGSAW network of univ... — <https://itnthub.tn.gov.in/technology-transfer.html>

8. iTNT Hub — Tamil Nadu Deep Tech Startup Policy (nodal a... — <https://itnthub.tn.gov.in/deeptechpolicy.html>

9. iTNT Hub — Tamil Nadu Deep Tech Startup Policy Book (TR... — <https://itnthub.tn.gov.in/assets/deep-tech-policy-book.pdf>

14.2 Support package requested

Support element	Why it is justified	Measurable output
Stage-gated early R&D; support	Funds evidence generation before revenue or certification is feasible.	G0–G3 artifacts, not sales volume.
Shared high-voltage, cell-simulation, HIL, thermal and EMC pre-compliance facility access	Removes the single largest capital barrier and raises laboratory safety discipline.	Booking and utilisation logs, calibration records, test reports.
Independent safety and cybersecurity mentorship (knowledge partner)	Challenges unsupported claims early and maps the correct approval route.	Signed design and compliance review minutes.
Test-agency orientation and pre-compliance guidance	Prevents wrong-standard and board-versus-pack certification errors.	Standards applicability matrix; test-readiness gap list.
IP protection and technology-transfer advisory	Protects local know-how and sequences filing correctly against public disclosure.	IP register; filing or trade-secret decision record.
Manufacturing-readiness and design-for-manufacture support	Moves from a one-off sample toward a reproducible quality process.	DFM/DFT, end-of-line and traceability plans; supplier risk assessment.
Facilitated, safety-gated integrator introductions	Needed only after G3 and controlled by safety gates.	Technical evaluation plans; no implied procurement.
Skills and talent support	Builds local embedded, validation, safety and quality capacity.	Named roles; training and certification outcomes.

14.3 Eligible cost categories

The Innovation Voucher Programme publishes the eligible cost categories reproduced below.¹⁶ The applicant's intended use of support maps onto them directly; amounts are deliberately not stated here and will be built from current quotations in the application's milestone and cost schedule.

Published eligible cost category ¹⁶	Intended programme use
Outsourcing charges, R&D;, consultancy, engineering design work	Independent functional-safety and cybersecurity review; specialist engineering design review; test-protocol development.
Raw materials / consumables / spares	Instrumented cells, connectors, harnessing, fixtures and consumables for the bench and A-sample.
Fabrication / manufacturing charges for POC and prototype development	A-sample fabrication and assembly under controlled release.
Intellectual property rights / patent filing	IP register establishment and filing actions arising from IP Clinic guidance.
Testing and validation / laboratory verification / certification	Bench and HIL campaigns, calibration services, EMC and environmental pre-compliance screening.
Commercialisation support services / consulting / feasibility studies / market-entry support	Structured customer discovery, manufacturing-readiness feasibility and localisation-method advisory.

14.4 Multiple funding asks are different capital envelopes

The applicant's earlier documents contain several different funding figures. These are not competing versions of one number; they are different capital envelopes with different scopes, sources and time horizons. The dossier states them as such, with no final rupee totals, because a credible schedule must be rebuilt from current quotations and the specific rules of each instrument.

Envelope	Scope it covers	Source category	Status
Envelope A — Voucher-scale evidence support	Independent review, prototype fabrication, testing and validation, IP actions, engineering design services	TNTTFC Innovation Voucher categories ¹⁶	This application

16. TNTTFC — Innovation Voucher Programme application (Vouc... — <https://www.techtransfer.itnthumb.tn.gov.in/innovation-voucherform/>

8. iTNT Hub — Tamil Nadu Deep Tech Startup Policy (nodal a... — <https://itnthumb.tn.gov.in/deeptechpolicy.html>

9. iTNT Hub — Tamil Nadu Deep Tech Startup Policy Book (TR... — <https://itnthumb.tn.gov.in/assets/deep-tech-policy-book.pdf>

Envelope	Scope it covers	Source category	Status
Envelope B — Early R&D; and test-bed support	Shared facility access, instrumentation time, extended bench campaign	State deep-tech instruments via the nodal agency ⁸⁹	Complementary; to be applied for separately if eligible
Envelope C — Formal compliance testing	Accredited testing and formal approval support for the integrated pack scope	Ring-fenced compliance budget, later gate	Deferred until the agency-confirmed scope exists
Envelope D — Industrialisation and working capital	Tooling, production quality systems, inventory, scale-up	Private capital and customer co-funding	Out of scope for public support

No final rupee total, quotation or price is asserted anywhere in this dossier. The financial schedule accompanying the application will be constructed from current supplier and laboratory quotations, will separate base cost, contingency with a stated risk basis, in-kind contribution and applicant co-funding, and will comply with the specific cost-category and contribution rules of the instrument applied for.

14.5 Milestone-linked disbursement principles

- Support is released against objective gate acceptance, not against narrative progress.
- Each tranche is tied to a named evidence deliverable from Section 11 and a gate from Section 10.2.
- Independent verification accompanies each safety-relevant or commercially material milestone.
- Grant-funded equipment is recorded in an asset register with a utilisation record.
- If a critical feasibility, safety or market assumption fails, funding stops or the scope is re-based. A documented no-go is an acceptable programme outcome.
- Excluded from any support request: mass-production inventory, sales travel, unbounded cloud expenditure, claims-based marketing, data-resale development, general corporate overhead beyond agreed project staffing, and any certification claim before results exist.

15. Localisation, DVA and Scheme-Linkage Evidence Plan

The PLI scheme for the automobile and auto-component industry applies a minimum 50% domestic value-addition condition to eligible approved products, with certification carried out through the notified process on an invoiced and costed bill-of-materials basis.⁴⁵⁶⁷ The applicant makes no DVA claim and no eligibility claim.

15.1 Wording control

Not stated in this dossier	Stated instead
"Our BMS has 65% DVA."	The programme will establish a costed bill of materials and a localisation plan; any domestic value-addition result will be determined under the applicable scheme methodology by the competent process. ⁵⁶
"It instantly makes OEMs PLI eligible."	A locally engineered component may support an eligible integrator's localisation strategy, subject to that integrator's scheme eligibility and product-level approval. ⁴⁷
"Imported BMS contributes 0% DVA."	Imported content treatment will be assessed under the applicable methodology; the programme is designed to increase local engineering, assembly, test and supply-chain content. ⁶
"PLI subsidy offsets the unit cost."	Commercial value from localisation is a hypothesis to be validated with an eligible applicant through the scheme process.
"Fully localized."	Progressive localisation of engineering, assembly, test and selected supply-chain content, measured against a documented method.

4. PIB — PLI Scheme for Automobile and Auto Components: ou... — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2287124®=3&lang=1>

5. PLI-Auto — Official Frequently Asked Questions (domesti... — https://pliauto.in/docs/guidelines/2021-12-22_FAQII.PDF

6. ACMA — Standard Operating Procedure for certification o... — https://www.acma.in/uploads/ciculer-attachement/PLI-Auto_DVASOP.pdf

7. PIB — PLI Scheme for Automobile sector: 50% domestic va... — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2290691®=3&lang=1>

15.2 DVA evidence plan

1. Define the legal manufacturer entity and the exact product scope to which any calculation would apply.
2. Freeze a costed bill of materials with overhead allocation and supplier-country data.
3. Maintain invoices, origin evidence and manufacturing-route records from first procurement.
4. Apply the published calculation methodology and standard operating procedure rather than an internally invented formula.⁶
5. Obtain professional or agency guidance before any percentage is used publicly.
6. Make no scheme-benefit claim in any material without confirmation for the specific applicant and approved product concerned.⁵

15.3 Wider policy context

PM E-DRIVE's phased manufacturing programme has been amended to require domestic manufacture of specified power-electronics assemblies for e-buses and e-trucks.¹¹¹² The applicant notes this as directional support for domestic traction-electronics capability. It is stated as context only: the applicant does not claim scheme participation, benefit or eligibility, and does not rely on any scheme outcome to justify the requested support.

16. Risk Register

Risk	Why it is material	Early indicator	Control	Gate consequence
Technical feasibility of the diagnostic measurement path at cell level	The central differentiator may require additional circuitry or may not scale	Supplier review or bench signal-quality assessment fails	Independent feasibility work package in WP1; alternative architecture path; narrowed claims	Stop or re-scope the advanced diagnostic scope
False negatives in hazard warning	Direct safety and liability exposure	Induced events missed in validation	Defence in depth; hard protections retained; conservative operating limits; extensive per-class validation	No authority-enabled pilot
Excess false alarms	Fleet downtime and loss of operator trust	Alerts per operating hour exceed the agreed target	Threshold governance in the restricted design record; confidence states; advisory-only operation; root-cause review	Remain in shadow mode
Thermal-state estimate unreliable across cell variants	Generalisation risk across suppliers and ageing states	High error on new supplier or aged cells	Physical sensors retained; per-variant validation; published validity envelope	Restrict the supported cell list
Compute, memory or latency shortfall	Proposed functions may not fit the target controller	Profiling exceeds the resource budget	Early resource profiling in WP1; scope optimisation; documented processor trade study	Revise architecture before fabrication
Certification scope misunderstanding	Funds could be committed to the wrong test	Test agency rejects a standalone subsystem scope	Written scoping with a competent agency at G0; correct AIS-038 Rev. 2 versus AIS-156 applicability analysis ¹²	Rebaseline the compliance budget
Safety and cybersecurity work underestimated	Automotive readiness delayed and rework accumulates	HARA, TARA or traceability missing at gate	Qualified assessor engaged from inception; work products explicitly budgeted	No pre-compliance or pilot authorisation
Component supply, lead time and obsolescence	Prototype and later production delay	Allocation notice or end-of-life notice	Lead-time register; approved alternates where feasible; lifecycle check; no safety-critical substitution without review	Adjust schedule; re-plan the fabrication tranche
Claims integrity	Inconsistent legacy figures could reach a reviewer	Any figure appearing without a label or protocol reference	Claims register (Section 18); single reconciled fact base; pre-publication review of every outward document	Withhold publication until reconciled
Schedule	Facility and reviewer availability drive the critical path	Booking slippage; gate dates missed	Early bookings; parallel work packages; explicit rebaseline at G0	Re-scope within the same gate structure

6. ACMA — Standard Operating Procedure for certification o... — https://www.acma.in/uploads/ciculer-attachement/PLI-Auto_DVASOP.pdf

5. PLI-Auto — Official Frequently Asked Questions (domesti... — https://pliauto.in/docs/guidelines/2021-12-22_FAQII.PDF

11. PIB — PM E-DRIVE scheme: outlay, e-bus support and phas... — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2290696®=3&lang=1>

12. <https://auto.economictimes.indiatimes.com/news/commercial-vehicle/government-amends-localisation-norms-for-e-buses-e-trucks-under-pm-e-drive-scheme/124268731>

1. PIB / MoRTH — Amendments to EV battery safety standards... — <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1862596®=3&lang=2>

2. PIB / MoRTH — Clarification on battery safety standards... — <https://www.pib.gov.in/PressReleaseDetail.aspx?PRID=2202970®=1&lang=1>

Risk	Why it is material	Early indicator	Control	Gate consequence
Partner and letter-of-intent evidence	Implied partnerships that are actually under discussion are easily falsified	Any draft referring to an unsigned party as a partner	Customer qualification ladder (Section 13.2); only executed dated documents are cited	Remove the reference before filing
Data protection and privacy	Fleet and vehicle data carry legal and trust obligations	Data collected without contractual authority	Opt-in, purpose limitation, retention limits, access control, separation from safety path (DAT-001 to DAT-003)	Fleet interface remains disabled
Cybersecurity	A connected controller expands the attack surface	Connectivity enabled before TARA completion	TARA before connectivity; secure boot and signed update; key management; SBOM; vulnerability process	Connectivity and OTA remain disabled
Budget and quotation risk	Estimates without current quotations are not auditable	Cost lines carried forward from older drafts	Rebuild every cost line from a current dated quotation; separate contingency with a stated risk basis	Withhold the tranche until re-quoted
IP leakage	A public application could expose differentiating know-how	Restricted material appearing in a public annex or shared drive	Three-tier disclosure model (Section 17); segregated document repository; NDA process; metadata stripping before submission	Suspend sharing and re-issue the sanitised version
Key-person and capacity risk	Safety-critical work cannot rest indefinitely on unpaid effort	Milestone slippage; review backlog	Realistic person-month budgeting; competency matrix; mentor support; documented succession of records	Re-plan scope to available capacity

17. Confidentiality, Disclosure Model and Rights

17.1 Three-tier disclosure model

Tier	Contents	Recipients and controls
Tier 1 — Public application (this document)	Problem, public value, use case, product boundary, black-box capabilities, external interfaces, target requirements with verification methods, safety and compliance pathway, validation strategy, work packages, gates, risks, governance and public outcomes.	Unrestricted for evaluation purposes. Assumed to be potentially disclosable if submitted to a public authority. Contains no protected material.
Tier 2 — Restricted technical annex (on written request, under NDA)	Deeper interface specification; selected architecture decisions and rationale; detailed test protocols; hazard and cybersecurity summaries; costed BOM ranges; non-public test result summaries and raw-data access terms.	Named evaluator only, under executed confidentiality terms, with watermarking, access logging, purpose limitation, no onward disclosure, a defined review window and return or destruction terms.
Tier 3 — Need-to-know controlled review (never routinely submitted)	Material indispensable to a specific assessment by a notified test agency, safety assessor or contracted integrator only.	Role-limited access, viewer log, expiry, watermark, no onward distribution, no external upload, prior IP and legal approval required.

17.2 Protected material excluded from all public disclosure

Protected category	Excluded completely	Safe public substitute used in this dossier
Algorithmic intellectual property	Algorithms, equations, model classes and structures, feature definitions, state variables, logic flows, thresholds, weights, training protocol, tuning and decision rules	"Proprietary on-device estimation and diagnostic module; interfaces, outputs and verification outcomes are documented."
Diagnostic sensing and signal method	Excitation or interrogation method, waveform and frequency characteristics, spectral treatment, mappings, internal-temperature derivation method, fault signatures and degradation indicators	"Supplementary internal thermal-state estimate and per-class abnormal-condition indication, validated against reference methods."
Software internals	Source or binary code, task scheduling, memory maps, control flow, compiler settings, debug traces, model export files, internal APIs	External interface control document, state and diagnostic definitions, released version identity.
Safety-sensitive configuration	Protection threshold values, contactor timing values, calibration maps and lookup tables, recovery strategies, fault-injection recipes, known weak points	Safety objectives, protection coverage categories, test categories and evidence summaries.

Protected category	Excluded completely	Safe public substitute used in this dossier
Cybersecurity material	Keys, certificates, seed and key logic, credentials, secure-boot configuration, vulnerability findings, attack paths	"Authenticated access and controlled update policy; detailed security evidence available under NDA if required."
Hardware design files	Schematics, Gerbers, PCB stack-up and layout, test-point maps, complete bill of materials, manufacturing files, enclosure detail	Public block-level architecture narrative, interface categories, non-confidential BOM categories.
Commercially sensitive material	Supplier quotations, detailed unit economics, customer pricing, valuations, negotiation terms, customer-specific parameter sets	High-level cost categories and the funding-need structure.
Personal and operational data	Vehicle identifiers, driver or fleet data, contact details, location histories, raw incident logs	Aggregated, de-identified KPI examples only.

17.3 Rights statement

All background intellectual property, inventions, know-how, proprietary methods, software, data models, test methods, design files and confidential information remain the exclusive property of the applicant or the relevant owner. Disclosure of this document grants no right to reproduce, reverse engineer, develop derivative works from, disclose, commercialise or otherwise use any material, except for the purpose of evaluating the requested support. Project reporting will provide objective evidence of milestone completion while preserving trade-secret boundaries. Nothing in this document constitutes a warranty, an offer, or a representation of fitness for any particular purpose.

17.4 What government reporting will and will not include

Will include: milestone completion certificates; third-party invoice and test evidence; redacted test reports showing protocol identifier, configuration, acceptance criterion and pass/fail result; an asset register and utilisation record for supported equipment; headcount, training and local-supplier indicators; change-control and risk-register extracts; and independent technical-review memoranda.

Will not include as routine grant reporting: unrestricted source-code escrow, mathematical formulations, raw proprietary diagnostic datasets, calibration maps, threshold values or manufacturing design packages.

17.5 Document handling guidance

- Public-filing artifacts are maintained in a repository separate from internal working material; internal deliberation notes, method documentation and design records are never co-located with submission packages.
- Document metadata is stripped before submission and file provenance is checked.
- Every outward technical sentence is checked against the Tier 1 boundary before release; if a sentence cannot be classified Tier 1, it is removed or rewritten.
- Public materials are kept deliberately non-enabling. Because India operates a first-to-file patent system, public disclosure sequencing is treated as an IP decision requiring counsel input, and IP Clinic support is requested for exactly this reason.¹⁵
- Any request for deeper detail is met with a Tier 2 or Tier 3 process, not with an ad-hoc disclosure.

15. TNTTFC — Startup programmes: Masterclass, IP Clinic, In... — <https://www.techtransfer.itnthumb.tn.gov.in/startup/>

18. Claims Register: High-Risk Source Wording and Approved Replacements

The applicant's earlier internal and pitch material contains wording that would not survive technical review. The register below records each high-risk formulation and the approved replacement used in this dossier. It is published deliberately: a reviewer should be able to see that the corrections were made systematically rather than selectively.

#	High-risk source claim	Why it fails	Approved replacement wording
1	"ASIL-D certified BMS"	Certification status misrepresentation; no assessment has been performed	Architecture intended to support an ISO 26262-aligned safety lifecycle using safety-capable components; formal assessment is future work.
2	"Completely prevents / eliminates thermal runaway"	Absolute safety claim; outcome depends on cells, pack, thermal design and operation	Designed to detect defined abnormal-condition precursors and initiate protective action before escalation, alongside independent protection mechanisms.
3	"Detects failing cells weeks in advance"	Untestable as stated and pooled across fault classes	Lead-time performance is a per-class validation objective; an early-warning objective of at least 48 hours applies only to defined scenarios and is pending validation.
4	"<1% SOC/SOH error"	Research-grade commitment for LFP across a full temperature envelope; conflicts with other internal figures	State-of-charge error not exceeding 2% as an A-sample target under a defined test matrix and reference method.
5	"Fault detection to contactor isolation under 100 microseconds"	Conflates electronic command with electromechanical actuation	Detection and command latency, and physical isolation completion, are specified, budgeted and verified as separate quantities.
6	"TRL 3/4 — validated in simulation"	No simulation artifact exists to inspect	Applicant-assessed early research stage, approximately TRL 2–3, subject to formal assessment through an inception evidence review.
7	"65% DVA / instantly unlocks PLI subsidies"	No audited costed BOM; scheme eligibility is applicant- and product-specific	A costed BOM and localisation plan will be produced; any DVA result is determined under the applicable methodology, and no scheme eligibility is conferred by the component. ⁵⁶
8	"AIS-156 certification for the BMS"	AIS-156 is the L-category standard; M and N category route is AIS-038 Rev. 2, and approval attaches to the vehicle or REESS, not a board	AIS-038 (Rev. 2) is the likely applicable traction-battery route for the M/N use case; final applicability is confirmed with the pack integrator and notified test agency. ¹²³
9	"India's first / only", "no Indian player offers..."	Unsupported superlative, contradicted by the applicant's own competitor notes	An intended differentiated capability combination; comparative statements will be made only from a dated, documented landscape analysis.
10	"Fully localized"	Absolute claim before any BOM audit	Progressive localisation of engineering, assembly, test and selected supply-chain content, measured against a documented method.
11	"Free forever SaaS saving Rs.20,000 per vehicle"	Monetisation and savings claim without customer validation or governance	An optional, non-safety, governed fleet interface is proposed; lifecycle cost and service model will be validated with customers.
12	"Eliminates thermistor blind spots / 100% coverage"	Unsourced precision and an implied removal of physical sensing	A supplementary internal thermal-state estimate that does not replace independent physical over-temperature protection unless supported by an approved safety case.
13	"Extends battery life by 30–40%"	No basis in any attached evidence	Removed. Life-extension is not claimed at product level; any future statement will cite published literature and be labelled as such.
14	"Our OEM relationships create a pipeline"	No executed document exists	Target accounts identified; no letter of intent, agreement, order or qualification is represented.
15	"Secured lab and manufacturing partner"	Facility access described elsewhere as under discussion	Facility and partner arrangements are described as targeted or under discussion until a dated executed document exists.
16	"Year 1 revenue of Rs.1 crore" and similar point forecasts	Presented as central estimates while the same corpus shows first revenue much later	Revenue is excluded from this technical dossier and held as scenario planning with stated assumptions in a separate financial annex.
17	"Instant PLI", "free forever", "guaranteed", "zero failure"	Absolutes that cannot be evidenced	Removed entirely. Conditional, testable formulations are used throughout.

5. PLI-Auto — Official Frequently Asked Questions (domestic... — https://pliauto.in/docs/guidelines/2021-12-22_FAQII.PDF

6. ACMA — Standard Operating Procedure for certification o... — https://www.acma.in/uploads/ciculer-attachement/PLI-Auto_DVASOP.pdf

1. PIB / MoRTH — Amendments to EV battery safety standards... — <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1862596®=3&lang=2>

2. PIB / MoRTH — Clarification on battery safety standards... — <https://www.pib.gov.in/PressReleaseDetail.aspx?PRID=2202970®=1&lang=1>

3. MoRTH — AIS-156 official standard text (approval of veh... — <https://morth.nic.in/sites/default/files/ASI/AIS-156.pdf>

13. PIB — MoRTH eDAR reply on EV accident and fire incident... — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2225866>

#	High-risk source claim	Why it fails	Approved replacement wording
18	EV fire counts presented as a growing safety crisis	The cited government reply also records that no centralised battery-fire dataset is maintained, and the series is small and flat	Severity- and evidence-based framing, with the data limitation stated explicitly. ¹³

19. Submission–Readiness Checklist and Conclusion

19.1 Accompanying items for the portal application

The portal's Innovation Voucher application requires specific applicant and project information.¹⁶ The checklist below lists what must be assembled. Nothing in this list is asserted to exist; each item must be attached only if it genuinely does.

Item required by the application	Preparation note
Applicant name, address, state, district, contact details, number of directors	Take from the registry record; must match the registration certificate exactly.
Company classification (Micro / Small / Medium Enterprise or DPIIT-certified startup)	Select only the classification the applicant can evidence.
DPIIT number and/or UAM number, and date of registration	From the certificate; the date must not be a future date.
UAM / MSME or Startup India registration certificate upload	Attach the actual certificate file.
Annual turnover (last year) and website, if applicable	State accurately; leave blank rather than estimate.
Voucher type selection (Voucher A up to Rs.3 lakh or Voucher B up to Rs.7 lakh)	Select against the scope and cost schedule actually proposed.
Project title, problem statement, solution proposed, detailed product and service description	Draw directly from Sections 2, 3, 5 and 6 of this dossier; keep within the character limit.
Invention or upgrade classification	State honestly with reference to the maturity statement in Section 4.
Scope of the project	Use Section 5.4 (in scope and out of scope).
Competitive and technical advantage	Use the differentiated-capability wording; no superlatives, no competitor disparagement.
Price advantage to customers	State as a provisional assumption pending the costed BOM; do not assert a price.
Total cost of the project and the milestone / activity cost table with applicant contribution	Build from current dated quotations; separate base cost, contingency and applicant contribution.
PPT presentation upload and video link	Prepare from this dossier; apply the same Tier 1 disclosure boundary to both.
Request for an iTNT knowledge partner	Requested — see Section 14.1.

19.2 Pre-submission checks

- Legal entity, registration particulars and Tamil Nadu operational presence are accurate and evidenced.
- One initial use case, vehicle category and pack configuration are selected and frozen.
- Every claim carries a Demonstrated / Planned / Target / Projection label.
- "Certified", "compliant", "first", "only", "eliminates", "prevents", "instant", "guarantee", "fully localized" and any DVA percentage have been removed unless documentary evidence is attached.
- AIS applicability is stated for the selected vehicle category and the pack or vehicle approval route is described correctly.
- Scheme wording is reviewed against the intended applicant, product and costed-BOM pathway.
- One staged, quotation-based cost schedule reconciles all cost lines, in-kind support and contingency.
- All named organisations are described as targeted unless a dated executed document exists.
- Data monetisation is excluded from the technical funding case.

16. TNTTFC — Innovation Voucher Programme application (Vouc... — <https://www.techtransfer.itnthumb.tn.gov.in/innovation-voucherform/>)

- The document has been screened for algorithms, equations, model detail, thresholds, keys, design files, supplier terms and personal data.
- The evidence repository and version-control owner are named.
- Independent reviewers are named or requested as a support deliverable.
- Document metadata is stripped and the submission bundle contains no internal working files.

19.3 Conclusion

The strongest case this applicant can make is not that the technology is finished, but that it is important, technically risky, measurable, locally valuable and governable. This dossier is written to invite scrutiny and to make that scrutiny affordable: one frozen use case; black-box, testable requirements; explicit unknowns; independent evidence gates; a conservative safety and cybersecurity posture; staged support tied to artifacts; quantified public outcomes stated as target types rather than invented numbers; a clear intellectual-property boundary; and honest commercial assumptions.

What is requested is support to convert a documented concept into independently evidenced engineering: shared test infrastructure, expert mentorship, IP guidance, TRL validation, pre-compliance orientation and milestone-linked early R&D; support through the technology-transfer pathway.¹⁴¹⁵¹⁶ What is offered in return is auditable evidence, reusable public validation knowledge, local engineering capability and skilled employment retained in Tamil Nadu — with proprietary methods protected and no safety, certification or economic claim made ahead of the evidence that would justify it.

Appendix A. Source Register

Every external factual statement in this document is cited inline by superscript number to the numbered footnote on the page where it appears. The complete register with full URLs follows. No internal file paths are cited anywhere in this document.

#	Source	URL
1	PIB / MoRTH — Amendments to EV battery safety standards AIS-156 and AIS-038 (Rev. 2), phased implementation	https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1862596@=3 =2
2	PIB / MoRTH — Clarification on battery safety standards applicable by vehicle category (AIS-156 for L-category; AIS-038 Rev. 2 for M and N categories)	https://www.pib.gov.in/PressReleaseDetail.aspx?PRID=2202970@=1 =1
3	MoRTH — AIS-156 official standard text (approval of vehicles / REESS)	https://morth.nic.in/sites/default/files/ASI/AIS-156.pdf
4	PIB — PLI Scheme for Automobile and Auto Components: outlay, DVA condition and certification status	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2287124@=3 =1
5	PLI-Auto — Official Frequently Asked Questions (domestic value addition and product approval process)	https://pliauto.in/docs/guidelines/2021-12-22_FAQII.PDF
6	ACMA — Standard Operating Procedure for certification of Domestic Value Addition under PLI-Auto	https://www.acma.in/uploads/ciculer-attachement/PLI-Auto_DVASOP.pdf
7	PIB — PLI Scheme for Automobile sector: 50% domestic value addition condition	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2290691@=3 =1
8	iTNT Hub — Tamil Nadu Deep Tech Startup Policy (nodal agency, R&D; support, test beds, commercialisation)	https://itnthub.tn.gov.in/deeptechpolicy.html
9	iTNT Hub — Tamil Nadu Deep Tech Startup Policy Book (TRL-banded support instruments)	https://itnthub.tn.gov.in/assets/deep-tech-policy-book.pdf
10	iTNT Hub — FOUNDATION Fund (eligibility and instrument terms)	https://itnthub.tn.gov.in/foundation_fund.html
11	PIB — PM E-DRIVE scheme: outlay, e-bus support and phased manufacturing programme	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2290696@=3 =1
12	ET Auto — Amended localisation norms for e-buses and e-trucks under PM E-DRIVE	https://auto.economictimes.indiatimes.com/news/commercial-vehicle/government-amends-localisation-norms-for-e-buses-e-trucks-under-pm-e-drive-scheme/124268731

14. TTTFC / iTNT Hub — Technology Transfer and Facilitatio... — <https://www.techtransfer.itnthub.tn.gov.in/>

15. TTTFC — Startup programmes: Masterclass, IP Clinic, In... — <https://www.techtransfer.itnthub.tn.gov.in/startup/>

16. TTTFC — Innovation Voucher Programme application (Vouc... — <https://www.techtransfer.itnthub.tn.gov.in/innovation-voucherform/>

#	Source	URL
13	PIB — MoRTH eDAR reply on EV accident and fire incident data (no centralised battery-fire dataset)	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2225866
14	TNTTFC / iTNT Hub — Technology Transfer and Facilitation Centre portal (Prototype to Product pathway)	https://www.techtransfer.itnthub.tn.gov.in/
15	TNTTFC — Startup programmes: Masterclass, IP Clinic, Innovation Spotlight, Technology Transfer, Technology Portfolio, Research2Revenue (Seed Fund at TRL 6+)	https://www.techtransfer.itnthub.tn.gov.in/startup/
16	TNTTFC — Innovation Voucher Programme application (Voucher A up to Rs.3 lakh, Voucher B up to Rs.7 lakh; eligible cost categories; knowledge-partner request)	https://www.techtransfer.itnthub.tn.gov.in/innovation-voucherform/
17	iTNT Hub — Technology Transfer / JIGSAW network of universities and researchers	https://itnthub.tn.gov.in/technology-transfer.html
18	iTNT Hub — Official website	https://itnthub.tn.gov.in/

Appendix B. Abbreviations

Term	Meaning
AIS	Automotive Industry Standard (India)
ASIL	Automotive Safety Integrity Level (ISO 26262)
BMS	Battery Management System
BOM	Bill of Materials
CAN / CAN-FD	Controller Area Network / CAN with Flexible Data-rate
DFM / DFT	Design for Manufacture / Design for Test
DPIIT	Department for Promotion of Industry and Internal Trade
DVA	Domestic Value Addition
EMC	Electromagnetic Compatibility
EOL	End-of-Line (production test)
FMEA / FTA	Failure Mode and Effects Analysis / Fault Tree Analysis
HARA	Hazard Analysis and Risk Assessment
HIL / MIL / SIL	Hardware-, Model- and Software-in-the-Loop testing
ICD	Interface Control Document
IVP	Innovation Voucher Programme
LFP	Lithium Iron Phosphate cell chemistry
MSME	Micro, Small and Medium Enterprise
OTA	Over-the-Air (software update)
PLI	Production Linked Incentive
R2R	Research2Revenue programme
REESS	Rechargeable Electrical Energy Storage System
SBOM	Software Bill of Materials
SOC / SOH	State of Charge / State of Health
TARA	Threat Analysis and Risk Assessment
TNTTFC	Tamil Nadu Technology Transfer and Facilitation Centre
TRL	Technology Readiness Level
UAM	Udyog Aadhaar Memorandum
WP	Work Package